MTH251 & MTH252

Mathematical Analysis I & II

Course Textbook

Department of Mathematical Sciences

UNIST

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Course Textbook

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Feel free to open issues or pull requests for corrections.

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Preface

Throughout this textbook, **logical formulas** are interpreted according to the following conventions:

- **Parentheses** have the highest precedence. For example,

  $$(P \vee Q) \wedge R$$

  is evaluated by first computing $P \vee Q$.

- The **logical connectives** are interpreted in the order

  $$\neg \; > \; \wedge \; > \; \vee \; > \implies > \iff$$

  Hence
  $$\neg P \wedge Q \vee R \implies S \iff T$$
  is understood as

  $$((((\neg P) \wedge Q) \vee R) \implies S) \iff T$$

- A **quantifier** binds the entire formula that follows it. Thus

  $$\forall x, P(x) \wedge Q(x) \implies R(x)$$

  means
  $$\forall x \,((P(x) \wedge Q(x)) \implies R(x))$$

- The symbol $\Downarrow$ is **not** a logical connective; it is used only to indicate the flow of a proof or an explanation. For example,

  $$\begin{gathered} P \\ \Downarrow \\ Q \end{gathered}$$

  merely expresses that P is true, and therefore Q is true in the current discussion; it is not itself a proposition.

Chapter 1

The Number Systems

1.1 The Natural Number Systems

The set $\{1, 2, 3, \ldots\}$ of **natural numbers** is denoted by $\mathbb{N}$, and is defined by the following axioms:

- $1 \in \mathbb{N}$
- $\forall n \in \mathbb{N}, s(n) \in \mathbb{N}$
- $\nexists n \in \mathbb{N}, s(n) = 1$
- $\forall m, n \in \mathbb{N}, s(m) = s(n) \implies m = n$
- If $S \subseteq \mathbb{N}$ satisfies
  - $1 \in S$
  - $\forall n \in S, s(n) \in S$

  then $S = \mathbb{N}$

where $s(n)$ is called the **successor** of n. These axioms are called the **Peano axioms** (or *Peano Postulates*).

Remark. The **Peano axioms** do not guarantee the *existence* of $\mathbb{N}$, which is guaranteed by the **axiom of infinity** in the **ZFC axioms**. In this textbook, the **ZFC axioms** is not discussed, so we will just assume that $\mathbb{N}$ exists.

Definition 1.1.1. Let $P(n)$ be a statement about $n \in \mathbb{N}$. Define

$$S := \{n \in \mathbb{N} \mid P(n) \text{ is true}\} \subseteq \mathbb{N}$$

If $P(n)$ satisfies

- $P(1)$ is true
- $\forall n \in \mathbb{N}, P(n)$ is true $\implies P(s(n))$ is true

Then S satisfies

- $1 \in S$
- $\forall n \in S, s(n) \in S$

Therefore,

$$\begin{gathered} S = \mathbb{N} \\ \Downarrow \\ \forall n \in \mathbb{N}, P(n) \text{ is true} \end{gathered}$$

This method of proof is called the **principle of mathematical induction**.

Definition 1.1.2. Let $n, m \in \mathbb{N}$ be given.

- The **addition** $+$ on $\mathbb{N}$ is defined as
  - $n + 1 := s(n)$
  - $n + s(m) := s(n + m)$
- The **multiplication** $\cdot$ on $\mathbb{N}$ is defined as
  - $n \cdot 1 := n$
  - $n \cdot s(m) := n \cdot m + n$

Remark. Strictly speaking, Definition 1.1.2 requires the **recursion theorem** to be strictly defined. In this textbook, the **recursion theorem** is not discussed, so we will just assume that the operations on $\mathbb{N}$ are **well-defined**.

Theorem 1.1.3. *Let $n, m, k \in \mathbb{N}$ be given. Then the following properties hold:*

- $n + m = m + n$
- $(n + m) + k = n + (m + k)$
- $n \cdot m = m \cdot n$
- $(n \cdot m) \cdot k = n \cdot (m \cdot k)$
- $n \cdot (m + k) = n \cdot m + n \cdot k$

Proof. We will not discuss the proof of this theorem, since it relies on the recursive definitions. □

Theorem 1.1.4. $\mathbb{N}$ *has the* ***well-ordering property****, which states that*

$$\forall S \subseteq \mathbb{N}, S \neq \varnothing \implies \exists m \in S, m = \min S$$

Proof. Assume that

$$\exists S \subseteq \mathbb{N}, S \neq \varnothing \wedge (\nexists m \in S, m = \min S)$$

Define

- $T := \mathbb{N} \setminus S \subseteq \mathbb{N}$
- $L := \{n \in \mathbb{N} \mid \{1, 2, \ldots, n\} \subseteq T\} \subseteq \mathbb{N}$

We will show that

$$\begin{gathered} L = \mathbb{N} \\ \Downarrow \\ T = \mathbb{N} \\ \Downarrow \\ S = \varnothing \\ \Downarrow \\ \perp \end{gathered}$$

- **Base case:** If $1 \in S$, then

  $$\begin{gathered} 1 \in S \\ \Downarrow \\ 1 = \min S \\ \Downarrow \\ \perp \end{gathered}$$

  Therefore,

  $$\begin{gathered} 1 \notin S \\ \Downarrow \\ 1 \in T \\ \Downarrow \\ 1 \in L \end{gathered}$$

- **Inductive step:** Let $n \in L$ be given. Then

  $$\begin{gathered} n \in L \\ \Downarrow \\ \{1, 2, \ldots, n\} \subseteq T \\ \Downarrow \\ \{1, 2, \ldots, n\} \cap S = \varnothing \end{gathered}$$

  If $n + 1 \in S$, then

  $$\begin{gathered} n + 1 \in S \\ \Downarrow \\ n + 1 = \min S \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\begin{gathered} n+1 \notin S \\ \Downarrow \\ n+1 \in T \\ \Downarrow \\ \{1,2,\ldots,n+1\} \subseteq T \\ \Downarrow \\ n+1 \in L \end{gathered}$$

By the *principle of mathematical induction*, $L = \mathbb{N}$. $\square$

1.2 Relations

Let S be a set. A **relation** $\sim$ on S is a subset of $S \times S$. Equivalently, a subset R of $S \times S$ is a relation on S. That is, for every $x, y \in S$,

- $R = \{(x,y) \in S \times S \mid x \sim y\}$
- $x \sim y \iff (x,y) \in R$

Definition 1.2.1. Let S be a set. A relation $\sim$ on S is called an **equivalence relation** if it satisfies the following properties:

- **Reflexivity**: $\forall x \in S, x \sim x$
- **Symmetry**: $\forall x, y \in S, x \sim y \implies y \sim x$
- **Transitivity**: $\forall x, y, z \in S, x \sim y \wedge y \sim z \implies x \sim z$

Definition 1.2.2. Let S be a set. A collection $\{A_\alpha\}_{\alpha \in I}$ of sets is called a **partition** of S if

- $\forall \alpha \in I, A_\alpha \neq \varnothing \wedge A_\alpha \subseteq S$
- $\forall \alpha, \beta \in I, \alpha \neq \beta \implies A_\alpha \cap A_\beta = \varnothing$
- $\displaystyle\bigcup_{\alpha \in I} A_\alpha = S$

Definition 1.2.3. Let S be a set, and let $\sim$ be an equivalence relation on S. For each $x \in S$, the **equivalence class** of x is defined as

$$[x] \coloneqq \{y \in S \mid y \sim x\}$$

We denote the set of all equivalence classes of S by

$$S/\sim \coloneqq \{[x] \mid x \in S\}$$

Theorem 1.2.4. *Let S be a set, and let $\sim$ be an equivalence relation on S. Then*

$$\forall x, y \in S, [x] \cap [y] \neq \varnothing \implies [x] = [y]$$

Proof. If $[x] \cap [y] \neq \varnothing$, then

$$\begin{gathered} \exists z \in S, z \in [x] \cap [y] \\ \Downarrow \\ z \in [x] \wedge z \in [y] \\ \Downarrow \\ z \sim x \wedge z \sim y \\ \Downarrow \\ x \sim z \wedge z \sim y \\ \Downarrow \\ x \sim y \end{gathered}$$

Therefore, for every $w \in [x]$,

$$\begin{gathered} w \sim x \wedge x \sim y \\ \Downarrow \\ w \sim y \\ \Downarrow \\ w \in [y] \end{gathered}$$

Therefore,

$$[x] \subseteq [y]$$

Similarly,

$$[y] \subseteq [x]$$

□

Theorem 1.2.5. *Let S be a set, and let $\{A_\alpha\}_{\alpha \in I}$ be a partition of S. Define*

$$x \sim y \iff \exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha$$

for all $x, y \in S$. Then

(A) *$\sim$ is an equivalence relation on S*

(B) *$\{A_\alpha\}_{\alpha \in I} = S/\sim$*

Proof. **(A)** Let $x, y, z \in S$ be given.

- **Reflexivity**: Since $x \in S$,

$$x \in S$$
$$\Downarrow$$
$$x \in \bigcup_{\alpha \in I} A_\alpha$$
$$\Downarrow$$
$$\exists \alpha \in I, x \in A_\alpha$$
$$\Downarrow$$
$$x \sim x$$

- **Symmetry**: If $x \sim y$, then

$$\exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha$$
$$\Downarrow$$
$$y \in A_\alpha \wedge x \in A_\alpha$$
$$\Downarrow$$
$$y \sim x$$

- **Transitivity**: If $x \sim y \wedge y \sim z$, then
  - $\exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha$
  - $\exists \beta \in I, y \in A_\beta \wedge z \in A_\beta$

  Therefore,

$$y \in A_\alpha \wedge y \in A_\beta$$
$$\Downarrow$$
$$A_\alpha \cap A_\beta \neq \varnothing$$
$$\Downarrow$$
$$\alpha = \beta$$
$$\Downarrow$$
$$x \in A_\alpha \wedge z \in A_\alpha$$
$$\Downarrow$$
$$x \sim z$$

(B) Let $x \in S$ be given. Then

$$x \in S$$
$$\Downarrow$$
$$x \in \bigcup_{\alpha \in I} A_\alpha$$
$$\Downarrow$$
$$\exists \alpha \in I, x \in A_\alpha$$

We will show that $A_\alpha = [x]$.

- ($\subseteq$) Let $y \in A_\alpha$ be given. Then

$$\begin{gathered} y \in A_\alpha \wedge x \in A_\alpha \\ \Downarrow \\ y \sim x \\ \Downarrow \\ y \in [x] \end{gathered}$$

- ($\supseteq$) Let $y \in [x]$ be given. Then

$$\begin{gathered} y \sim x \\ \Downarrow \\ \exists \beta \in I, y \in A_\beta \wedge x \in A_\beta \\ \Downarrow \\ x \in A_\alpha \wedge x \in A_\beta \\ \Downarrow \\ x \in A_\alpha \cap A_\beta \\ \Downarrow \\ A_\alpha \cap A_\beta \neq \varnothing \\ \Downarrow \\ \alpha = \beta \\ \Downarrow \\ y \in A_\alpha \end{gathered}$$

To sum up,

$$\begin{gathered} \forall x \in S, \exists \alpha \in I, A_\alpha = [x] \\ \Downarrow \\ S/\sim \subseteq \{A_\alpha\}_{\alpha \in I} \end{gathered}$$

Also,

$$\begin{gathered} \forall \alpha \in I, A_\alpha \neq \varnothing \\ \Downarrow \\ \exists x \in A_\alpha \subseteq S \\ \Downarrow \\ A_\alpha = [x] \\ \Downarrow \\ \{A_\alpha\}_{\alpha \in I} \subseteq S/\sim \end{gathered}$$

Therefore,

$$\{A_\alpha\}_{\alpha \in I} = S/\sim$$

□

1.3 The Integer and Rational Number Systems

Define a relation $\sim$ on $\mathbb{N} \times \mathbb{N}$ as

$$(a, b) \sim (c, d) \iff a + d = b + c$$

Let $(a, b), (c, d), (e, f) \in \mathbb{N} \times \mathbb{N}$ be given. By Theorem 1.1.3,

- **Reflexivity**: Since $a \in \mathbb{N} \wedge b \in \mathbb{N}$,

$$\begin{gathered} a + b = b + a \\ \Downarrow \\ (a, b) \sim (a, b) \end{gathered}$$

- **Symmetry**: If $(a, b) \sim (c, d)$, then

$$\begin{gathered} a + d = b + c \\ \Downarrow \\ c + b = d + a \\ \Downarrow \\ (c, d) \sim (a, b) \end{gathered}$$

- **Transitivity**: If $(a, b) \sim (c, d) \wedge (c, d) \sim (e, f)$, then

$$\begin{gathered} a + d = b + c \wedge c + f = d + e \\ \Downarrow \\ a + d + f = b + c + f = b + d + e \\ \Downarrow \\ a + f = b + e \\ \Downarrow \\ (a, b) \sim (e, f) \end{gathered}$$

Therefore, $\sim$ is an equivalence relation on $\mathbb{N} \times \mathbb{N}$. The set of **integers** $\mathbb{Z}$ is defined as

$$\mathbb{Z} := (\mathbb{N} \times \mathbb{N}) / \sim$$

Definition 1.3.1. Let $[(a, b)], [(c, d)] \in \mathbb{Z}$ be given.

- The **addition** $+$ on $\mathbb{Z}$ is defined as

$$[(a, b)] + [(c, d)] := [(a + c, b + d)]$$

- The **multiplication** $\cdot$ on $\mathbb{Z}$ is defined as

$$[(a, b)] \cdot [(c, d)] := [(a \cdot c + b \cdot d, a \cdot d + b \cdot c)]$$

Remark. The operations $+$ and $\cdot$ are **well-defined** on $\mathbb{Z}$; that is, they do not depend on the **choice of representatives** of the equivalence classes. Indeed, if

- $(a, b) \sim (a', b')$

- $(c, d) \sim (c', d')$

then one can verify that

- $(a + c, b + d) \sim (a' + c', b' + d')$
- $(a \cdot c + b \cdot d, a \cdot d + b \cdot c) \sim (a' \cdot c' + b' \cdot d', a' \cdot d' + b' \cdot c')$

The same issue arises for **addition** and **multiplication** on $\mathbb{Q}$, whose **well-definedness** could be verified in exactly the same manner.

Theorem 1.3.2. *Let $n, m, k \in \mathbb{Z}$ be given. Then the following properties hold:*

- $n + m = m + n$
- $(n + m) + k = n + (m + k)$
- $n \cdot m = m \cdot n$
- $(n \cdot m) \cdot k = n \cdot (m \cdot k)$
- $n \cdot (m + k) = (n \cdot m) + (n \cdot k)$

Proof. The proof follows directly from Definition 1.3.1 and theorem 1.1.3. □

Define
$$\mathbb{Z}^+ := \{[(n + 1, 1)] \in \mathbb{Z} \mid n \in \mathbb{N}\} \subseteq \mathbb{Z}$$
Define a relation $\sim$ on $\mathbb{Z} \times \mathbb{Z}^+$ as

$$(a, b) \sim (c, d) \iff a \cdot d = b \cdot c$$

Let $(a, b), (c, d), (e, f) \in \mathbb{Z} \times \mathbb{Z}^+$ be given. By Theorem 1.3.2,

- **Reflexivity**: Since $a \in \mathbb{Z} \wedge b \in \mathbb{Z}^+$,

$$\begin{gathered} a \cdot b = b \cdot a \\ \Downarrow \\ (a, b) \sim (a, b) \end{gathered}$$

- **Symmetry**: If $(a, b) \sim (c, d)$, then

$$\begin{gathered} a \cdot d = b \cdot c \\ \Downarrow \\ c \cdot b = d \cdot a \\ \Downarrow \\ (c, d) \sim (a, b) \end{gathered}$$

- **Transitivity**: If $(a, b) \sim (c, d) \wedge (c, d) \sim (e, f)$, then

$$\begin{gathered} a \cdot d = b \cdot c \wedge c \cdot f = d \cdot e \\ \Downarrow \\ a \cdot d \cdot f = b \cdot c \cdot f = b \cdot d \cdot e \\ \Downarrow \\ a \cdot f = b \cdot e \\ \Downarrow \\ (a, b) \sim (e, f) \end{gathered}$$

Therefore, $\sim$ is an equivalence relation on $\mathbb{Z} \times \mathbb{Z}^+$. The set of **rational numbers** $\mathbb{Q}$ is defined as

$$\mathbb{Q} \coloneqq (\mathbb{Z} \times \mathbb{Z}^+)/\sim$$

Definition 1.3.3. Let $[(a, b)], [(c, d)] \in \mathbb{Q}$ be given.

- The **addition** $+$ on $\mathbb{Q}$ is defined as

$$[(a, b)] + [(c, d)] \coloneqq [(ad + bc, bd)]$$

- The **multiplication** $\cdot$ on $\mathbb{Q}$ is defined as

$$[(a, b)] \cdot [(c, d)] \coloneqq [(ac, bd)]$$

Theorem 1.3.4. *Let $p, q, r \in \mathbb{Q}$ be given. Then the following properties hold:*

- $p + q = q + p$
- $(p + q) + r = p + (q + r)$
- $p \cdot q = q \cdot p$
- $(p \cdot q) \cdot r = p \cdot (q \cdot r)$
- $p \cdot (q + r) = (p \cdot q) + (p \cdot r)$

Proof. The proof follows directly from Definition 1.3.3 and theorem 1.3.2. □

Remark. Strictly speaking, the constructions of $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ produce **different kind** of sets. For example,

- $n \in \mathbb{N}$
- $[(n + 1, 1)] \in \mathbb{Z}$

- $[(n, [(2, 1)])] \in \mathbb{Q}$

are distinct objects in the underlying set-theoretic construction. Nevertheless, there are **canonical embeddings**

- $\mathbb{N} \xrightarrow{\cong} \mathbb{Z}$ such that $n \mapsto [(n + 1, 1)]$
- $\mathbb{Z} \xrightarrow{\cong} \mathbb{Q}$ such that $a \mapsto [(a, [(2, 1)])]$

which preserve the **algebraic operations**. We therefore identify each number system with its image in the next larger one and write

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$$

This identification is justified by the fact that the embeddings are canonical, requiring **no arbitrary choices**. Throughout this textbook, inclusions such as

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

should be understood as canonical identifications rather than literal inclusions arising directly from the set-theoretic constructions.

Remark. Define

$$0 \coloneqq [(1, 1)] \in \mathbb{Z}$$

In this textbook, we will use the following notations:

- $\mathbb{N}_0 \coloneqq \mathbb{N} \cup \{0\} \subseteq \mathbb{Z}$
- $\mathbb{N}_n \coloneqq \{k \in \mathbb{N} \mid k \geq n\} = \{n, n + 1, n + 2, \ldots\} \subseteq \mathbb{N}$ for each $n \in \mathbb{N}$
- $\mathbb{N}^0 \coloneqq \varnothing$
- $\mathbb{N}^m \coloneqq \{k \in \mathbb{N} \mid k \leq m\} = \{1, 2, \ldots, m\} \subseteq \mathbb{N}$ for each $m \in \mathbb{N}$
- $\mathbb{N}_n^m \coloneqq \{k \in \mathbb{N}_n \mid k \leq m\} = \{n, n + 1, \ldots, m\} \subseteq \mathbb{N}_0$ for each $n, m \in \mathbb{N}_0$ with $n \leq m$

1.4 Ordered Sets

A set S is called an **ordered set** if it is equipped with a relation $<$ such that for every $x, y, z \in S$, the following properties hold:

- $x < y \wedge y < z \implies x < z$
- Only one of the relations $x < y$, $x = y$, or $x > y$ holds

The relation $<$ is called an **(total) order** on S.

Definition 1.4.1. The relations $>$, $\leq$, and $\geq$ are defined as

- $x > y \iff y < x$
- $x \geq y \iff x > y \lor x = y$
- $x \leq y \iff y \geq x$

Example 1.4.2. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are ordered sets with the following relations:

- $\forall n, m \in \mathbb{N}, m < n \iff \exists k \in \mathbb{N}, n = m + k$
- $\forall n, m \in \mathbb{Z}, m < n \iff \exists k \in \mathbb{Z}^+, n = m + k$
- $\forall p, q \in \mathbb{Q}, q < p \iff \exists r \in \mathbb{Q}^+, p = q + r$

where

$$\mathbb{Q}^+ \coloneqq \left\{[(p, q)] \in \mathbb{Q} \;\middle|\; p, q \in \mathbb{Z}^+\right\} \subseteq \mathbb{Q}$$

Definition 1.4.3. Let S be an ordered set and $E \subseteq S$.

- E is said to be **bounded above** if

  $$\exists M \in S, \forall x \in E, x \leq M$$

  Such an M is called an **upper bound** of E.
- E is said to be **bounded below** if

  $$\exists m \in S, \forall x \in E, m \leq x$$

  Such an m is called a **lower bound** of E.

Theorem 1.4.4. *Let $E \subseteq \mathbb{Z}$ be nonempty. Then*

(A) *E is bounded below $\implies$ E has a least element*

(B) *E is bounded above $\implies$ E has a greatest element*

Proof. **(A)** Let E be bounded below. Then,

$$\begin{gathered} E \text{ is bounded below} \\ \Downarrow \\ \exists m \in \mathbb{Z}, \forall x \in E, m \leq x \end{gathered}$$

Define

$$F \coloneqq \{x - m + 1 \mid x \in E\} \subseteq \mathbb{N}$$

Then

$$\begin{gathered} E \neq \varnothing \\ \Downarrow \\ F \neq \varnothing \end{gathered}$$

By Theorem 1.1.4,

$$\begin{gathered} \exists n \in F, n = \min F \\ \Downarrow \\ \forall x \in E, n \leq x - m + 1 \\ \Downarrow \\ \forall x \in E, n + m - 1 \leq x \end{gathered}$$

Since $n = \min F$,

$$\begin{gathered} n \in F \\ \Downarrow \\ \exists x_0 \in E, n = x_0 - m + 1 \\ \Downarrow \\ x_0 = n + m - 1 = \min E \end{gathered}$$

(B) Let E be bounded above. Then,

$$\begin{gathered} E \text{ is bounded above} \\ \Downarrow \\ \exists M \in \mathbb{Z}, \forall x \in E, x \leq M \end{gathered}$$

Define

$$F := \{-x + M + 1 \mid x \in E\} \subseteq \mathbb{N}$$

Then

$$\begin{gathered} E \neq \varnothing \\ \Downarrow \\ F \neq \varnothing \end{gathered}$$

By Theorem 1.1.4,

$$\begin{gathered} \exists n \in F, n = \min F \\ \Downarrow \\ \forall x \in E, n \leq -x + M + 1 \\ \Downarrow \\ \forall x \in E, x \leq M - n + 1 \end{gathered}$$

Since $n = \min F$,

$$\begin{gathered} n \in F \\ \Downarrow \\ \exists x_0 \in E, n = -x_0 + M + 1 \\ \Downarrow \\ x_0 = M - n + 1 = \max E \end{gathered}$$

$\square$

Definition 1.4.5. Let S be an ordered set, and let $E \subseteq S$ be bounded above. An element $\alpha \in S$ is called a **supremum** (or *least upper bound*) of E if

- α is an upper bound of E
- $\forall \beta \in S, \beta < \alpha \implies \beta$ is not an upper bound of E

A supremum of E is denoted by

$$\alpha = \sup E$$

Similarly, let $E \subseteq S$ be bounded below. An element $\alpha \in S$ is called an **infimum** (or *greatest lower bound*) of E if

- α is a lower bound of E
- $\forall \beta \in S, \beta > \alpha \implies \beta$ is not a lower bound of E

An infimum of E is denoted by

$$\alpha = \inf E$$

Example 1.4.6. Define

$$S := \left\{ p \in \mathbb{Q}^+ \;\middle|\; p^2 < 2 \right\}$$

- Let $p \in S$ be given. Define
$$q := p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} > p$$
Then
$$\begin{gathered} q^2 - 2 = \frac{2(p^2 - 2)}{(p+2)^2} < 0 \\ \Downarrow \\ q \in S \end{gathered}$$
This implies that $p \in S$ is not an upper bound of S.
- Let $p \in \mathbb{Q}^+ \setminus S$ be given.
$$\nexists q \in \mathbb{Q}^+, q^2 = 2$$
Therefore,
$$\begin{gathered} p^2 \geq 2 \\ \Downarrow \\ p^2 > 2 \\ \Downarrow \\ \forall x \in S, x^2 < 2 < p^2 \\ \Downarrow \\ \forall x \in S, x < p \\ \Downarrow \\ p \text{ is an upper bound of } S \end{gathered}$$

Define

$$q := p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} < p$$

Then

$$\begin{gathered} q^2 - 2 = \frac{2(p^2 - 2)}{(p+2)^2} > 0 \\ \Downarrow \\ q^2 > 2 \\ \Downarrow \\ \forall x \in S, x^2 < 2 < q^2 \\ \Downarrow \\ \forall x \in S, x < q \\ \Downarrow \\ q \text{ is an upper bound of } S \end{gathered}$$

This implies that p is not a least upper bound of S.

To sum up,

$$\nexists \alpha \in \mathbb{Q}^+, \alpha = \sup S$$

Definition 1.4.7. Let S be an ordered set and $E \subseteq S$. If

- $E \neq \varnothing$
- E is bounded above

implies that

$$\exists \alpha \in S, \alpha = \sup E$$

then S is said to have the **least upper bound property**.

Theorem 1.4.8. *Let S be an ordered set and $E \subseteq S$. Then*

- *S has the least upper bound property*
- *$E \neq \varnothing$*
- *E is bounded below*

implies that

$$\exists \alpha \in S, \alpha = \inf E$$

Proof. Assume that

- S has the *least upper bound property*
- $E \neq \varnothing$

- E is bounded below

Define
$$L := \{\lambda \in S \mid \forall x \in E, \lambda \leq x\} \subseteq S$$

Then
$$\begin{gathered} E \text{ is bounded below} \\ \Downarrow \\ L \neq \varnothing \end{gathered}$$

Also,
$$\begin{gathered} E \neq \varnothing \\ \Downarrow \\ \exists \beta \in E \\ \Downarrow \\ \forall \lambda \in L, \lambda \leq \beta \\ \Downarrow \\ L \text{ is bounded above} \end{gathered}$$

By the *least upper bound property*,
$$\exists \alpha \in S, \alpha = \sup L$$

We will show that
$$\alpha = \inf E$$

Assume that $\alpha \notin L$. Then
$$\begin{gathered} \alpha \notin L \\ \Downarrow \\ \exists \beta \in E, \beta < \alpha \end{gathered}$$

Since $\alpha = \sup L$,
$$\begin{gathered} \alpha = \sup L \\ \Downarrow \\ \beta \text{ is not an upper bound of } L \\ \Downarrow \\ \exists \gamma \in L, \beta < \gamma \leq \alpha \end{gathered}$$

Then
$$\begin{gathered} \gamma \in L \\ \Downarrow \\ \forall x \in E, \gamma \leq x \\ \Downarrow \\ \gamma \leq \beta \\ \Downarrow \\ \beta < \gamma \leq \beta \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\alpha \in L$$
$$\Downarrow$$
$$\alpha \text{ is a lower bound of } E$$

Since $\alpha = \sup L$,

$$\forall \delta \in S, \delta > \alpha \implies \delta \notin L$$

To sum up,

- α is a lower bound of E
- $\forall \delta \in S, \delta > \alpha \implies \delta$ is not a lower bound of E

Therefore,

$$\alpha = \inf E$$

□

1.5 Fields and Ordered Fields

A set F is called a **field** if it is equipped with two binary operations

- **addition** $+$
- **multiplication** $\cdot$

and satisfies the following axioms:

(A) Axioms for **addition**

- $\forall x, y \in F, x + y \in F$
- $\forall x, y \in F, x + y = y + x$
- $\forall x, y, z \in F, (x + y) + z = x + (y + z)$
- $\exists 0 \in F, \forall x \in F, x + 0 = x$
- $\forall x \in F, \exists -x \in F, x + (-x) = 0$

(B) Axioms for **multiplication**

- $\forall x, y \in F, x \cdot y \in F$
- $\forall x, y \in F, x \cdot y = y \cdot x$
- $\forall x, y, z \in F, (x \cdot y) \cdot z = x \cdot (y \cdot z)$
- $\exists 1 \in F, 1 \neq 0 \wedge (\forall x \in F, x \cdot 1 = x)$
- $\forall x \in F, x \neq 0 \implies \exists x^{-1} \in F, x \cdot x^{-1} = 1$

(C) The **distributive law**

- $\forall x, y, z \in F, x \cdot (y + z) = x \cdot y + x \cdot z$

A set F is called an **ordered field** if

- F is a field
- F is an ordered set

and satisfies the following axioms:

- $\forall x, y, z \in F, x < y \implies x + z < y + z$
- $\forall x, y \in F, x > 0 \land y > 0 \implies x \cdot y > 0$

Example 1.5.1. $(\mathbb{Q}, +, \cdot)$ is an ordered field, while $(\mathbb{Z}, +, \cdot)$ is not an ordered field since

$$\nexists 2^{-1} \in \mathbb{Z}, 2 \cdot 2^{-1} = 1$$

1.6 The Real Number System

An ordered field $(\mathbb{R}, +, \cdot)$ has the *least upper bound property*. The construction of $\mathbb{R}$ from $\mathbb{Q}$ is rather complicated, so it will be discussed separately in Appendix. In this section, we will just assume that $\mathbb{R}$ exists.

Theorem 1.6.1. *$\mathbb{R}$ has the **Archimedean property**, which states that*

$$\forall x \in \mathbb{R}^+, \forall y \in \mathbb{R}, \exists n \in \mathbb{N}, nx > y$$

where

$$\mathbb{R}^+ := \{x \in \mathbb{R} \mid x > 0\}$$

Proof. Define

$$E := \{nx \mid nx \leq y,\ n \in \mathbb{N}\} \subseteq \mathbb{R}$$

If $E = \varnothing$, then

$$\begin{gathered} 1 \in \mathbb{N} \land 1 \cdot x \notin E \\ \Downarrow \\ 1 \cdot x > y \end{gathered}$$

Otherwise,

$$E \neq \varnothing$$

E is bounded above by y. By the *least upper bound property*,

$$\exists \alpha \in \mathbb{R}, \alpha = \sup E$$

Since $\alpha = \sup E$,

$$\begin{gathered}\alpha - x < \alpha \\ \Downarrow \\ \alpha - x \text{ is not an upper bound of } E \\ \Downarrow \\ \exists m \in \mathbb{N}, mx > \alpha - x \\ \Downarrow \\ (m+1)x > \alpha \\ \Downarrow \\ m+1 \in \mathbb{N} \wedge (m+1)x \notin E \\ \Downarrow \\ (m+1)x > y\end{gathered}$$

Therefore, we can take $n = m + 1$ to complete the proof. □

Corollary 1.6.2.

$$\forall \varepsilon \in \mathbb{R}^+, \forall b \in (1, +\infty), \exists n \in \mathbb{N}, b^{-n} < \varepsilon$$

Proof. The **Bernoulli's inequality** states that

$$\forall x \in [-1, +\infty), \forall n \in \mathbb{N}, (1+x)^n \geq 1 + nx$$

By Theorem 1.6.1,

$$\begin{gathered}\forall a \in \mathbb{R}^+, \exists n \in \mathbb{N}, na > \frac{1}{\varepsilon} \\ \Downarrow \\ (1+a)^n \geq 1 + na > \frac{1}{\varepsilon} \implies (1+a)^{-n} < \varepsilon\end{gathered}$$

Then

$$\begin{gathered}b \in (1, +\infty) \\ \Downarrow \\ b - 1 \in \mathbb{R}^+ \\ \Downarrow \\ \exists n \in \mathbb{N}, n(b-1) > \frac{1}{\varepsilon} \\ \Downarrow \\ (1 + (b-1))^n \geq 1 + n(b-1) > \frac{1}{\varepsilon} \implies (1 + (b-1))^{-n} < \varepsilon \\ \Downarrow \\ b^{-n} < \varepsilon\end{gathered}$$

□

Theorem 1.6.3. *Let $E \subseteq \mathbb{Z}$ be nonempty. Then*

(A) *E is bounded below in $\mathbb{R}$ $\implies$ E has a least element*

(B) *E is bounded above in $\mathbb{R}$ $\implies$ E has a greatest element*

Proof. **(A)** Let E be bounded below in $\mathbb{R}$. Then

$$\exists \alpha \in \mathbb{R}, \forall x \in E, \alpha \leq x$$

If $\alpha \geq 0$, then

$$\begin{gathered} \forall x \in E, 0 \leq \alpha \leq x \\ \Downarrow \\ 0 \in \mathbb{Z} \text{ is a lower bound of } E \end{gathered}$$

Otherwise, by Theorem 1.6.1,

$$\begin{gathered} -\alpha \in \mathbb{R} \\ \Downarrow \\ \exists n \in \mathbb{N}, n > -\alpha \\ \Downarrow \\ \forall x \in E, -n < \alpha \leq x \\ \Downarrow \\ -n \in \mathbb{Z} \text{ is a lower bound of } E \end{gathered}$$

Therefore, E is bounded below in $\mathbb{Z}$. By Theorem 1.4.4, E has a least element.

(B) Let E be bounded above in $\mathbb{R}$. Then

$$\exists \alpha \in \mathbb{R}, \forall x \in E, x \leq \alpha$$

Then

$$\begin{gathered} \alpha \in \mathbb{R} \\ \Downarrow \\ \exists n \in \mathbb{N}, n > \alpha \\ \Downarrow \\ \forall x \in E, x \leq \alpha < n \\ \Downarrow \\ n \in \mathbb{Z} \text{ is an upper bound of } E \end{gathered}$$

Therefore, E is bounded above in $\mathbb{Z}$. By Theorem 1.4.4, E has a greatest element.

□

Definition 1.6.4. Let $x \in \mathbb{R}$ be given. The **floor function** $\lfloor x \rfloor$ and **ceiling function** $\lceil x \rceil$ are defined as

- $\lfloor x \rfloor := \max\{k \in \mathbb{Z} \mid k \leq x\}$

- $\lceil x \rceil := \min\{k \in \mathbb{Z} \mid k \geq x\}$

Theorem 1.6.3 guarantees the *existence* of $\lfloor x \rfloor$ and $\lceil x \rceil$.

Theorem 1.6.5. *$\mathbb{Q}$ is **dense** in $\mathbb{R}$, which states that*

$$\forall x, y \in \mathbb{R}, x < y \implies \exists p \in \mathbb{Q}, x < p < y$$

Proof. Define

$$\varepsilon := y - x > 0$$

By Theorem 1.6.1,

$$\exists n \in \mathbb{N}, \frac{1}{n} < \varepsilon$$

Define

$$S := \left\{ k \in \mathbb{Z} \;\middle|\; \frac{k}{n} \leq x \right\} \subseteq \mathbb{Z}$$

Then

$$\begin{gathered} \lfloor x \rfloor \leq x < \lfloor x \rfloor + 1 \\ \Downarrow \\ n\lfloor x \rfloor \leq nx < n\lfloor x \rfloor + n \\ \Downarrow \\ n\lfloor x \rfloor \in S \\ \Downarrow \\ S \neq \varnothing \end{gathered}$$

S is bounded above by nx. By Theorem 1.6.3,

$$\exists m \in \mathbb{Z}, m = \max S$$

Since $m = \max S$,

$$\begin{gathered} m \in S \\ \Downarrow \\ \frac{m}{n} \leq x \\ \Downarrow \\ \frac{m+1}{n} \leq x + \frac{1}{n} \end{gathered}$$

Also,

$$\begin{gathered} m + 1 \notin S \\ \Downarrow \\ \frac{m+1}{n} > x \end{gathered}$$

To sum up,

$$x < \frac{m+1}{n} \leq x + \frac{1}{n} < x + \varepsilon = x + (y - x) = y$$

Therefore, we can take $p = \dfrac{m+1}{n} \in \mathbb{Q}$ to complete the proof. $\square$

Theorem 1.6.6.

$$\forall x \in \mathbb{R}^+, \forall n \in \mathbb{N}, \exists! y \in \mathbb{R}^+, y^n = x$$

Such y is denoted by

- $y = \sqrt[n]{x}$
- $y = x^{\frac{1}{n}}$

Proof. Define

$$E \coloneqq \left\{t \in \mathbb{R}^+ \;\middle|\; t^n < x\right\}$$

If $x \geq 1$, then

$$\begin{gathered} \forall m \in \mathbb{N}, 2^{-m} < 1 \leq x \\ \Downarrow \\ 2^{-1} \in E \\ \Downarrow \\ E \neq \varnothing \end{gathered}$$

Otherwise,

$$\begin{gathered} x < 1 \\ \Downarrow \\ \forall m \in \mathbb{N}, x^{2m} < x \\ \Downarrow \\ x^2 \in E \\ \Downarrow \\ E \neq \varnothing \end{gathered}$$

E is bounded above by max$\{x, 1\}$. By the *least upper bound property*,

$$\exists \alpha \in \mathbb{R}^+, \alpha = \sup E$$

We will show that $\alpha^n = x$. Let $0 < a < b < \infty$ be given. Then

$$b^n - a^n = (b-a)\sum_{k=0}^{n-1} b^{n-1-k}a^k < (b-a)nb^{n-1}$$

Assume that $\alpha^n < x$. By Theorem 1.6.5,

$$\exists h \in \mathbb{R}^+, h < \min\left\{1, \frac{x - \alpha^n}{n(\alpha+1)^{n-1}}\right\}$$

Put $a = \alpha$ and $b = \alpha + h$. Then

$$\begin{gathered} (\alpha+h)^n - \alpha^n < hn(\alpha+h)^{n-1} < hn(\alpha+1)^{n-1} \\ \Downarrow \\ (\alpha+h)^n < \alpha^n + hn(\alpha+1)^{n-1} < \alpha^n + (x - \alpha^n) = x \end{gathered}$$

Since $\alpha = \sup E$,

$$\begin{gathered}(\alpha + h)^n < x \\ \Downarrow \\ \alpha + h \in E \\ \Downarrow \\ \perp\end{gathered}$$

Assume that $\alpha^n > x$. By Theorem 1.6.5,

$$\exists h \in \mathbb{R}^+, h < \min\left\{\alpha, \frac{\alpha^n - x}{n\alpha^{n-1}}\right\}$$

Put $a = \alpha - h$ and $b = \alpha$. Then

$$\begin{gathered}\alpha^n - (\alpha - h)^n < hn\alpha^{n-1} \\ \Downarrow \\ (\alpha - h)^n > \alpha^n - hn\alpha^{n-1} > \alpha^n - (\alpha^n - x) = x\end{gathered}$$

Since $\alpha = \sup E$,

$$\begin{gathered}(\alpha - h)^n > x \\ \Downarrow \\ \forall t \in E, t^n < x < (\alpha - h)^n \\ \Downarrow \\ \forall t \in E, t < \alpha - h \\ \Downarrow \\ \alpha - h \text{ is an upper bound of } E \\ \Downarrow \\ \perp\end{gathered}$$

Therefore,

$$\alpha^n = x$$

This guarantees the *existence* of y. Let $0 < \alpha < \beta < \infty$ be given. Then

$$\begin{gathered}\beta^n - \alpha^n = (\beta - \alpha)\sum_{k=0}^{n-1} \beta^{n-1-k}\alpha^k > 0 \\ \Downarrow \\ \beta^n > \alpha^n\end{gathered}$$

This guarantees the *uniqueness* of y. □

Definition 1.6.7. Let $y \in [0, 1)$ and $b \in \mathbb{N}_2$ be given. A sequence $(a_n)_{n=1}^{\infty}$ is called a **greedy base b expansion** of y if

$$\forall n \in \mathbb{N}, 0 \leq y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n}$$

where

$$\forall k \in \mathbb{N}, a_k \in S_b := \{0, 1, 2, \ldots, b-1\}$$

In particular, if $b = 10$, such a sequence is called a **greedy decimal expansion** of y.

Theorem 1.6.8. *Let $y \in [0, 1)$ and $b \in \mathbb{N}_2$ be given. Then the greedy base b expansion of y exists and is unique.*

Proof. Define

$$\begin{array}{ccc} t_1 := by & T_1 := \{t \in S_b \mid t \leq t_1\} & a_1 := \max T_1 \\ t_2 := b^2 y - b a_1 & T_2 := \{t \in S_b \mid t \leq t_2\} & a_2 := \max T_2 \\ \vdots & \vdots & \vdots \\ t_n := b^n y - \displaystyle\sum_{k=1}^{n-1} a_k b^{n-k} & T_n := \{t \in S_b \mid t \leq t_n\} & a_n := \max T_n \end{array}$$

Let $m \in \mathbb{N}$ be given. If $0 \leq t_m < b$, then

- $0 \in T_m$
- T_m is bounded above by $b - 1$

By Theorem 1.4.4,

$$\begin{gathered} \exists M \in S_b, M = \max T_m \\ \Downarrow \\ a_m = M \text{ is well-defined} \end{gathered}$$

Define $P(n)$:

- $0 \leq t_n < b$
- $\forall k \in \mathbb{N}^n, a_k = \max T_k$ is well-defined

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Since $0 \leq t_1 = by < b$,
  - $0 \in T_1$
  - T_1 is bounded above by $b - 1$

  By Theorem 1.4.4,

$$\begin{gathered} a_1 = \max T_1 \text{ is well-defined} \\ \Downarrow \\ P(1) \text{ is true} \end{gathered}$$

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Then

$$\begin{aligned} t_{n+1} &= b^{n+1}y - \sum_{k=1}^{n} a_k b^{n+1-k} \\ &= b\left(b^n y - \sum_{k=1}^{n} a_k b^{n-k}\right) \\ &= b\left(b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} - a_n\right) \\ &= b(t_n - a_n) \end{aligned}$$

By the inductive hypothesis,

- $0 \le t_n < b$
- $a_n = \max T_n$

Then

$$\begin{gathered} a_n \le t_n < a_n + 1 \\ \Downarrow \\ 0 \le t_n - a_n < 1 \\ \Downarrow \\ 0 \le t_{n+1} = b(t_n - a_n) < b \\ \Downarrow \\ P(n+1) \text{ is true} \end{gathered}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. Then

$$\begin{gathered} a_n \le b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} < a_n + 1 \\ \Downarrow \\ 0 \le b^n y - \sum_{k=1}^{n} a_k b^{n-k} < 1 \\ \Downarrow \\ 0 \le y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n} \end{gathered}$$

for all $n \in \mathbb{N}$. This guarantees the *existence* of a *greedy base b expansion* of y. Let two sequences $(a_n)_{n=1}^{\infty}$ and $(c_n)_{n=1}^{\infty}$ satisfy

- $\forall n \in \mathbb{N}, 0 \le y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n}$
- $\forall n \in \mathbb{N}, 0 \le y - \sum_{k=1}^{n} \frac{c_k}{b^k} < \frac{1}{b^n}$

where

$$\forall k \in \mathbb{N}, a_k \in S_b \wedge c_k \in S_b$$

We will show that

$$\forall n \in \mathbb{N}, a_n = c_n$$

Define

$$Q(n) : \forall k \in \mathbb{N}^n, a_k = c_k$$

We will show that $Q(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** By the assumption,

$$\begin{gathered} 0 \le y - \frac{a_1}{b} < \frac{1}{b} \\ \Downarrow \\ \frac{a_1}{b} \le y < \frac{a_1+1}{b} \end{gathered} \qquad \begin{gathered} 0 \le y - \frac{c_1}{b} < \frac{1}{b} \\ \Downarrow \\ \frac{c_1}{b} \le y < \frac{c_1+1}{b} \end{gathered}$$

By combining the two inequalities,

$$\begin{gathered} \frac{a_1}{b} \le y < \frac{c_1+1}{b} \\ \Downarrow \\ a_1 < c_1 + 1 \\ \Downarrow \\ a_1 \le c_1 \end{gathered} \qquad \begin{gathered} \frac{c_1}{b} \le y < \frac{a_1+1}{b} \\ \Downarrow \\ c_1 < a_1 + 1 \\ \Downarrow \\ c_1 \le a_1 \end{gathered}$$

Therefore,

$$\begin{gathered} a_1 = c_1 \\ \Downarrow \\ Q(1) \text{ is true} \end{gathered}$$

- **Inductive step:** Let $Q(n)$ be true for $n \in \mathbb{N}$. Then

$$\begin{gathered} 0 \le y - \sum_{k=1}^{n+1} \frac{a_k}{b^k} < \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{a_k}{b^k} \le y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} \end{gathered} \qquad \begin{gathered} 0 \le y - \sum_{k=1}^{n+1} \frac{c_k}{b^k} < \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{c_k}{b^k} \le y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} \end{gathered}$$

By combining the two inequalities,

$$\begin{gathered} \sum_{k=1}^{n+1} \frac{a_k}{b^k} \le y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{a_k - c_k}{b^k} < \frac{1}{b^{n+1}} \end{gathered} \qquad \begin{gathered} \sum_{k=1}^{n+1} \frac{c_k}{b^k} \le y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{c_k - a_k}{b^k} < \frac{1}{b^{n+1}} \end{gathered}$$

By the inductive hypothesis,

$$\forall k \in \mathbb{N}^n, a_k = c_k$$

Then

$$\begin{array}{ccc} \dfrac{a_{n+1} - c_{n+1}}{b^{n+1}} < \dfrac{1}{b^{n+1}} & \qquad & \dfrac{c_{n+1} - a_{n+1}}{b^{n+1}} < \dfrac{1}{b^{n+1}} \\ \Downarrow & & \Downarrow \\ a_{n+1} - c_{n+1} < 1 & & c_{n+1} - a_{n+1} < 1 \\ \Downarrow & & \Downarrow \\ a_{n+1} \leq c_{n+1} & & c_{n+1} \leq a_{n+1} \end{array}$$

Therefore,

$$\begin{array}{c} a_{n+1} = c_{n+1} \\ \Downarrow \\ Q(n+1) \text{ is true} \end{array}$$

By the *principle of mathematical induction*, $Q(n)$ is true for all $n \in \mathbb{N}$. This guarantees the *uniqueness* of the *greedy base b expansion* of y. □

Theorem 1.6.9. *Let $b \in \mathbb{N}_2$ be given. Let*

- $(a_n)_{n=1}^{\infty}$ *be the greedy base b expansion of $a \in [0, 1)$*
- $(c_n)_{n=1}^{\infty}$ *be the greedy base b expansion of $c \in [0, 1)$*

Then

$$(\forall n \in \mathbb{N}, a_n = c_n) \implies a = c$$

Proof. By Corollary 1.6.2, if $a \neq c$ (*WLOG* $a < c$), then

$$\exists n \in \mathbb{N}, \frac{1}{b^n} < c - a$$

By Definition 1.6.7,

$$\begin{array}{ccc} 0 \leq a - \displaystyle\sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n} & \qquad & 0 \leq c - \displaystyle\sum_{k=1}^{n} \frac{c_k}{b^k} < \frac{1}{b^n} \\ \Downarrow & & \Downarrow \\ \displaystyle\sum_{k=1}^{n} \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} & & c < \displaystyle\sum_{k=1}^{n} \frac{c_k}{b^k} + \frac{1}{b^n} \end{array}$$

By the assumption,

$$\begin{gathered}\forall n \in \mathbb{N}, a_n = c_n \\ \Downarrow \\ \sum_{k=1}^{n} \frac{a_k}{b^k} = \sum_{k=1}^{n} \frac{c_k}{b^k} \\ \Downarrow \\ \sum_{k=1}^{n} \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} < c < \sum_{k=1}^{n} \frac{c_k}{b^k} + \frac{1}{b^n} \\ \Downarrow \\ \perp\end{gathered}$$

Therefore,

$$a = c$$

□

Theorem 1.6.10. *Let $b \in \mathbb{N}_2$ be given. Let $(a_n)_{n=1}^{\infty}$ be such that*

$$\forall k \in \mathbb{N}, a_k \in S_b$$

If

$$\nexists y \in [0,1), (a_n)_{n=1}^{\infty} \text{ is the greedy base } b \text{ expansion of } y$$

then

$$\exists N \in \mathbb{N}, \forall n \in \mathbb{N}_N, a_n = b - 1$$

Proof. We will prove the contrapositive: If

$$\forall N \in \mathbb{N}, \exists n \in \mathbb{N}_N, a_n \neq b - 1$$

then

$$\exists y \in [0,1), (a_n)_{n=1}^{\infty} \text{ is the } \textit{greedy base } b \textit{ expansion} \text{ of } y$$

Assume that

$$\forall N \in \mathbb{N}, \exists n \in \mathbb{N}_N, a_n \neq b - 1$$

Define

$$x_n := \sum_{k=1}^{n} \frac{a_k}{b^k}$$

for each $n \in \mathbb{N}$. Then

$$\begin{gathered}\forall k \in \mathbb{N}, a_k \in S_b \\ \Downarrow \\ x_n = \sum_{k=1}^{n} \frac{a_k}{b^k} \leq \sum_{k=1}^{n} \frac{b-1}{b^k} = 1 - \frac{1}{b^n} < 1\end{gathered}$$

for all $n \in \mathbb{N}$. Therefore, $\{x_n\}_{n=1}^{\infty}$ is bounded above by 1. By the *least upper bound property*,

$$\exists y \in \mathbb{R}, y = \sup\{x_n\}_{n=1}^{\infty}$$

First, we will show that $y \in [0, 1)$. By the assumption,

$$\exists m \in \mathbb{N}, a_m \neq b - 1$$

Then

$$x_n = x_m + \sum_{k=m+1}^{n} \frac{a_k}{b^k} \leq x_m + \sum_{k=m+1}^{n} \frac{b-1}{b^k}$$
$$\Downarrow$$
$$x_n \leq x_m + \frac{1}{b^m} - \frac{1}{b^n} < x_m + \frac{1}{b^m}$$

for all $n > m$. Since $a_m \leq b - 2$,

$$\begin{aligned} x_m + \frac{1}{b^m} &= \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{a_m + 1}{b^m} \\ &\leq \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{b-1}{b^m} \\ &\leq \sum_{k=1}^{m} \frac{b-1}{b^k} = 1 - \frac{1}{b^m} \end{aligned}$$

To sum up,

- $\forall n \in \mathbb{N}, n \leq m \implies x_n \leq x_m$
- $\forall n \in \mathbb{N}, n > m \implies x_n < x_m + \dfrac{1}{b^m} \leq 1 - \dfrac{1}{b^m}$

Therefore, $1 - \dfrac{1}{b^m}$ is an upper bound of $\{x_n\}_{n=1}^{\infty}$. Since $0 \leq x_1 \leq y$,

$$y = \sup\{x_n\}_{n=1}^{\infty}$$
$$\Downarrow$$
$$y \leq 1 - \frac{1}{b^m} < 1$$
$$\Downarrow$$
$$y \in [0, 1)$$

Next, we will show that $(a_n)_{n=1}^{\infty}$ is the *greedy base b expansion* of y. Let $n \in \mathbb{N}$ be fixed. By the assumption,

$$\exists m \in \mathbb{N}_{n+1}, a_m \leq b - 2$$

Then

$$\begin{gathered} x_m = x_n + \sum_{k=n+1}^{m} \frac{a_k}{b^k} < x_n + \sum_{k=n+1}^{m} \frac{b-1}{b^k} \\ \Downarrow \\ x_m < x_n + \frac{1}{b^n} - \frac{1}{b^m} \\ \Downarrow \\ x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n} \end{gathered}$$

By the same argument as above, $x_m + \dfrac{1}{b^m}$ is an upper bound of $\{x_n\}_{n=1}^{\infty}$. Then

$$\begin{gathered} y = \sup\{x_n\}_{n=1}^{\infty} \\ \Downarrow \\ y \leq x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n} \\ \Downarrow \\ 0 \leq y - x_n < \frac{1}{b^n} \end{gathered}$$

Since n was arbitrary, $(a_n)_{n=1}^{\infty}$ is the *greedy base* b *expansion* of y. □

Theorem 1.6.11. *Let $b \in \mathbb{N}_2$ be given. Let $(a_n)_{n=1}^{\infty}$ be such that*

$$\forall k \in \mathbb{N}, a_k \in S_b$$

If

$$\exists N \in \mathbb{N}, \forall n \in \mathbb{N}_N, a_n = b - 1$$

then

$$\nexists y \in [0,1), (a_n)_{n=1}^{\infty} \textit{ is the greedy base } b \textit{ expansion of } y$$

This theorem is the ***converse*** *of Theorem 1.6.10.*

Proof. Assume that

$$\exists N \in \mathbb{N}, \forall n \in \mathbb{N}_N, a_n = b - 1$$

If

$$\exists y \in [0,1), (a_n)_{n=1}^{\infty} \text{ is the } \textit{greedy base } b \textit{ expansion} \text{ of } y$$

then

$$0 \leq y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n}$$

for all $n \in \mathbb{N}$. Define

$$x_n := \sum_{k=1}^{n} \frac{a_k}{b^k}$$

for each $n \in \mathbb{N}$. Then

$$x_n = x_N + \sum_{k=N+1}^{n} \frac{b-1}{b^k} = x_N + \frac{1}{b^N} - \frac{1}{b^n}$$
$$\Downarrow$$
$$y - x_n = y - x_N - \frac{1}{b^N} + \frac{1}{b^n} \geq 0$$
$$\Downarrow$$
$$\frac{1}{b^n} \geq x_N + \frac{1}{b^N} - y$$

for all $n > N$. Since $N \in \mathbb{N}$,

$$0 \leq y - x_N < \frac{1}{b^N}$$
$$\Downarrow$$
$$x_N + \frac{1}{b^N} - y > 0$$

By Corollary 1.6.2,

$$\exists m \in \mathbb{N}, \frac{1}{b^m} < x_N + \frac{1}{b^N} - y$$
$$\Downarrow$$
$$\forall n \in \mathbb{N}, n > \max\{m, N\} \implies \frac{1}{b^n} < x_N + \frac{1}{b^N} - y$$
$$\Downarrow$$
$$\perp$$

Therefore,

$$\nexists y \in [0,1), (a_n)_{n=1}^{\infty} \text{ is the } \textit{greedy base } b \textit{ expansion} \text{ of } y$$

□

1.7 The Extended Real Number System

The Extended real number system $\overline{\mathbb{R}}$ is defined as

$$\overline{\mathbb{R}} \coloneqq \mathbb{R} \cup \{-\infty, +\infty\}$$

We define the following operations on $\overline{\mathbb{R}}$:

- $\forall x \in \mathbb{R} \cup \{+\infty\}, x + (+\infty) \coloneqq +\infty$
- $\forall x \in \mathbb{R} \cup \{-\infty\}, x + (-\infty) \coloneqq -\infty$
- $\forall x \in \mathbb{R}^+ \cup \{+\infty\}, x \cdot (\pm\infty) \coloneqq \pm\infty$
- $\forall x \in \mathbb{R}^- \cup \{-\infty\}, x \cdot (\pm\infty) \coloneqq \mp\infty$

- $\forall x \in \mathbb{R}, \dfrac{x}{\pm\infty} := 0$

The following operations are undefined in $\overline{\mathbb{R}}$:

- $(\pm\infty) + (\mp\infty)$
- $0 \cdot (\pm\infty)$
- $\dfrac{(\pm\infty)}{(\pm\infty)}$ and $\dfrac{(\pm\infty)}{(\mp\infty)}$
- $\dfrac{x}{0}$ for all $x \in \overline{\mathbb{R}}$

$\overline{\mathbb{R}}$ is not a field, but it is an ordered set with the following order relation:

$$\forall x \in \mathbb{R}, -\infty < x < +\infty$$

Remark. We can identify the following intervals in $\overline{\mathbb{R}}$:

- $\forall a \in \mathbb{R}, (-\infty, a) = \{x \in \mathbb{R} \mid x < a\}$
- $\forall a \in \mathbb{R}, (a, +\infty) = \{x \in \mathbb{R} \mid x > a\}$
- $(-\infty, +\infty) = \mathbb{R}$.

Theorem 1.7.1. *For all $E \subseteq \overline{\mathbb{R}}$,*

- $\exists x \in \overline{\mathbb{R}}, x = \sup E$
- $\exists y \in \overline{\mathbb{R}}, y = \inf E$

Proof. If $E = \varnothing$, then

- $\sup E = -\infty$
- $\inf E = +\infty$

Otherwise,

- If E is bounded above in $\mathbb{R}$, then
$$\exists x \in \mathbb{R}, x = \sup E$$
by the *least upper bound property*.
- Otherwise, $\sup E = +\infty$

Also,

- If E is bounded below in $\mathbb{R}$, then
$$\exists y \in \mathbb{R}, y = \inf E$$
by Theorem 1.4.8.
- Otherwise, $\inf E = -\infty$

$\square$

1.8 The Complex Number System

The complex number system $\mathbb{C}$ is defined as

$$\mathbb{C} := \{(x, y) \mid x, y \in \mathbb{R}\}$$

The binary operations on $\mathbb{C}$ are defined as

- $\forall x, y, u, v \in \mathbb{R}, (x, y) + (u, v) := (x + u, y + v)$
- $\forall x, y, u, v \in \mathbb{R}, (x, y) \cdot (u, v) := (xu - yv, xv + yu)$

One can verify that $\mathbb{C}$ is a field with the following properties:

- The **additive identity** is $(0, 0)$
- The **multiplicative identity** is $(1, 0)$
- The **additive inverse** of (x, y) is $(-x, -y)$
- The **multiplicative inverse** of (x, y) is

$$\left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2}\right)$$

  for $(x, y) \neq (0, 0)$

The **imaginary unit** i is defined as

$$i := (0, 1) \in \mathbb{C}$$

Then

$$i^2 = (0, 1) \cdot (0, 1) = (-1, 0)$$

There is a **canonical embedding**

$$\mathbb{R} \xrightarrow{\cong} \mathbb{C} \text{ such that } x \mapsto (x, 0)$$

Therefore, we identify $\mathbb{R}$ with its image in $\mathbb{C}$ and write

$$\mathbb{R} \subseteq \mathbb{C}$$

Then, for every complex number $(x, y) \in \mathbb{C}$,

$$(x, y) = (x, 0) + (0, y) = (x, 0) + (y, 0) \cdot (0, 1) = x + yi$$

Definition 1.8.1. Let $z = (x, y) \in \mathbb{C}$ be given.

- The **real part** of z is defined as

  - $\Re(z) := x$
  - $\operatorname{Re}(z) := x$
- The **imaginary part** of z is defined as
  - $\Im(z) := y$
  - $\operatorname{Im}(z) := y$
- The **complex conjugate** of z is defined as

$$\overline{z} := (x, -y) = x - yi$$

Theorem 1.8.2. *Let $z = (x, y) \in \mathbb{C}$ and $w = (u, v) \in \mathbb{C}$ be given. Then the following properties hold:*

- $\overline{(\overline{z})} = z$
- $z = \overline{z} \iff z \in \mathbb{R}$
- $\overline{z + w} = \overline{z} + \overline{w}$
- $\overline{zw} = \overline{z} \cdot \overline{w}$
- $z + \overline{z} = 2\operatorname{Re}(z)$
- $z - \overline{z} = 2i\operatorname{Im}(z)$
- $z\overline{z} = x^2 + y^2 \in \mathbb{R}^+ \cup \{0\}$

Proof. The proof follows directly from Definition 1.8.1. □

Definition 1.8.3. Let $z = (x, y) \in \mathbb{C}$ be given. The **modulus** of z is defined as

$$|z| := \sqrt{z\overline{z}} = \sqrt{x^2 + y^2} \in \mathbb{R}^+ \cup \{0\}$$

The *existence* and *uniqueness* of $|z|$ are guaranteed by Theorems 1.6.6 and 1.8.2

Theorem 1.8.4. *Let $a_1, a_2, \ldots, a_n \in \mathbb{C}$ and $b_1, b_2, \ldots, b_n \in \mathbb{C}$ be given. Then*

$$\left| \sum_{k=1}^{n} a_k \overline{b_k} \right|^2 \leq \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2$$

This theorem is called the ***Cauchy-Schwarz inequality****.*

Proof. Let $1 \le i < j \le n$ be given. Then

$$\begin{aligned} |a_i b_j - a_j b_i|^2 &= (a_i b_j - a_j b_i)\overline{(a_i b_j - a_j b_i)} \\ &= |a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2 - (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}) \end{aligned}$$

Consider the following sums:

$$\begin{aligned} \sum_{i<j} (|a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2) &= \sum_{i<j} (|a_i|^2 |b_j|^2) + \sum_{j<i} (|a_i|^2 |b_j|^2) \\ &= \sum_{i \ne j} (|a_i|^2 |b_j|^2) \\ &= \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2 - \sum_{k=1}^{n} |a_k|^2 |b_k|^2 \end{aligned}$$

$$\begin{aligned} \sum_{i<j} (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}) &= \sum_{i \ne j} (a_i \overline{b_i} \cdot \overline{a_j} b_j) \\ &= \sum_{i=1}^{n} a_i \overline{b_i} \cdot \sum_{j=1}^{n} \overline{a_j} b_j - \sum_{k=1}^{n} a_k \overline{b_k} \cdot \overline{a_k} b_k \\ &= \left| \sum_{k=1}^{n} a_k \overline{b_k} \right|^2 - \sum_{k=1}^{n} |a_k|^2 |b_k|^2 \end{aligned}$$

Therefore,

$$\sum_{i<j} |a_i b_j - a_j b_i|^2 = \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2 - \left| \sum_{k=1}^{n} a_k \overline{b_k} \right|^2 \ge 0$$

□

Corollary 1.8.5. *Let $a_1, a_2, \ldots, a_n \in \mathbb{R}$ and $b_1, b_2, \ldots, b_n \in \mathbb{R}$ be given. Then*

$$\left(\sum_{k=1}^{n} a_k b_k \right)^2 \le \sum_{k=1}^{n} a_k^2 \cdot \sum_{k=1}^{n} b_k^2$$

Theorem 1.8.6. *Let $z, w \in \mathbb{C}$ be given. Then the following properties hold:*

(A) $|zw| = |z| \cdot |w|$

(B) $|z + w| \le |z| + |w|$

(C) $||z| - |w|| \le |z - w|$

Proof. Let $z = (x, y) \in \mathbb{C}$ and $w = (u, v) \in \mathbb{C}$ be given.

(A) $|zw|^2 = (xu - yv)^2 + (xv + yu)^2 = (x^2 + y^2)(u^2 + v^2) = |z|^2 \cdot |w|^2$

(B) By Corollary 1.8.5,

$$(xu + yv)^2 \leq (x^2 + y^2)(u^2 + v^2)$$

Then

$$\begin{aligned}|z + w|^2 &= (x + u)^2 + (y + v)^2 \\ &= x^2 + y^2 + u^2 + v^2 + 2(xu + yv) \\ &\leq x^2 + y^2 + u^2 + v^2 + 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| + |w|)^2\end{aligned}$$

(C) Again, by Corollary 1.8.5,

$$\begin{aligned}|z - w|^2 &= (x - u)^2 + (y - v)^2 \\ &= x^2 + y^2 + u^2 + v^2 - 2(xu + yv) \\ &\geq x^2 + y^2 + u^2 + v^2 - 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| - |w|)^2\end{aligned}$$

□

1.9 Euclidean Spaces

For each $n \in \mathbb{N}$, the **Euclidean n-space** $\mathbb{R}^n$ is defined as

$$\mathbb{R}^n := \{\boldsymbol{x} = (x_1, x_2, \ldots, x_n) \mid \forall k \in \mathbb{N}^n, x_k \in \mathbb{R}\}$$

where $x_1, x_2, \ldots, x_n$ are called the **coordinates** of $\boldsymbol{x}$. The elements of $\mathbb{R}^n$ for $n > 1$ are called **points** (or *vectors*) and are represented by bold symbols.

The operations on $\mathbb{R}^n$ are defined as

- $\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \boldsymbol{x} + \boldsymbol{y} := (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)$
- $\forall \alpha \in \mathbb{R}, \forall \boldsymbol{x} \in \mathbb{R}^n, \alpha\boldsymbol{x} := (\alpha x_1, \alpha x_2, \ldots, \alpha x_n)$

Definition 1.9.1. Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ be given.

- The **inner product** of $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined as

$$\boldsymbol{x} \cdot \boldsymbol{y} := \sum_{k=1}^{n} x_k y_k$$

- The **norm** of $\boldsymbol{x}$ is defined as

$$\|\boldsymbol{x}\| := \sqrt{\boldsymbol{x} \cdot \boldsymbol{x}} = \sqrt{\sum_{k=1}^{n} x_k^2}$$

- The **distance** between $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined as

$$d(\boldsymbol{x}, \boldsymbol{y}) := \|\boldsymbol{x} - \boldsymbol{y}\| = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}$$

where one can verify that $d(x, y) = |x - y|$ in the case of $n = 1$.

Theorem 1.9.2. *Let $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^n$ be given. Then the following properties hold:*

(A) $|\boldsymbol{x} \cdot \boldsymbol{y}| \leq \|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\|$

(B) $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$

(C) $\|\boldsymbol{x} - \boldsymbol{y}\| \geq |\|\boldsymbol{x}\| - \|\boldsymbol{y}\||$

Proof. **(A)** By Corollary 1.8.5,

$$|\boldsymbol{x} \cdot \boldsymbol{y}|^2 = \left| \sum_{k=1}^{n} x_k y_k \right|^2 \leq \sum_{k=1}^{n} x_k^2 \cdot \sum_{k=1}^{n} y_k^2 = \|\boldsymbol{x}\|^2 \cdot \|\boldsymbol{y}\|^2$$

(B) By part **(A)**,

$$\begin{aligned} \|\boldsymbol{x} + \boldsymbol{y}\|^2 &= (\boldsymbol{x} + \boldsymbol{y}) \cdot (\boldsymbol{x} + \boldsymbol{y}) = \|\boldsymbol{x}\|^2 + 2\boldsymbol{x} \cdot \boldsymbol{y} + \|\boldsymbol{y}\|^2 \\ &\leq \|\boldsymbol{x}\|^2 + 2|\boldsymbol{x} \cdot \boldsymbol{y}| + \|\boldsymbol{y}\|^2 \leq \|\boldsymbol{x}\|^2 + 2\|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\| + \|\boldsymbol{y}\|^2 \\ &= (\|\boldsymbol{x}\| + \|\boldsymbol{y}\|)^2 \end{aligned}$$

(C) By part **(B)**,

$$\begin{gathered} \|\boldsymbol{x}\| = \|(\boldsymbol{x} - \boldsymbol{y}) + \boldsymbol{y}\| \leq \|\boldsymbol{x} - \boldsymbol{y}\| + \|\boldsymbol{y}\| \\ \Downarrow \\ \|\boldsymbol{x}\| - \|\boldsymbol{y}\| \leq \|\boldsymbol{x} - \boldsymbol{y}\| \end{gathered}$$

Also,

$$\begin{gathered} \|\boldsymbol{y}\| = \|(\boldsymbol{y} - \boldsymbol{x}) + \boldsymbol{x}\| \leq \|\boldsymbol{y} - \boldsymbol{x}\| + \|\boldsymbol{x}\| \\ \Downarrow \\ \|\boldsymbol{y}\| - \|\boldsymbol{x}\| \leq \|\boldsymbol{x} - \boldsymbol{y}\| \end{gathered}$$

To sum up,

$$\|\boldsymbol{x} - \boldsymbol{y}\| \geq |\|\boldsymbol{x}\| - \|\boldsymbol{y}\||$$

□

Chapter 2

Basic Topology

2.1 Functions

A **function** (or *mapping*) is defined with the following three components:

- A **domain** X
- A **codomain** Y
- A **rule** f which assigns a *unique* element $y \in Y$ to each $x \in X$

f is said to be a function from X to Y, which is denoted by

$$f : X \to Y$$

For each $x \in X$, the *unique* element $y \in Y$ assigned to x is denoted by

$$y = f(x)$$

Definition 2.1.1. Let $f : X \to Y$ be given. For every $E \subseteq X$, the **image** of E under f is defined as

$$f(E) \coloneqq \{f(x) \mid x \in E\} \subseteq Y$$

In particular, if $E = X$, then $f(X) \subseteq Y$ is called the **range** of f.

Definition 2.1.2. Let $f : X \to Y$ be given. For every $E \subseteq Y$, the **inverse image** of E under f is defined as

$$f^{-1}(E) \coloneqq \{x \in X \mid f(x) \in E\} \subseteq X$$

Definition 2.1.3. Let $f : X \to Y$ be given.

- f is **injective** (or *one-to-one*) if

$$\forall x_1, x_2 \in X, f(x_1) = f(x_2) \implies x_1 = x_2$$

- f is **surjective** (or *onto*) if

$$\forall y \in Y, \exists x \in X, f(x) = y$$

or equivalently,

$$f(X) = Y$$

- f is **bijective** if it is both injective and surjective

Remark. The following notations are used to denote the properties of functions:

- $f : X \hookrightarrow Y$ denotes that f is injective
- $f : X \twoheadrightarrow Y$ denotes that f is surjective
- $f : X \xrightarrow{\sim} Y$ or $f : X \xrightarrow{\cong} Y$ denotes that f is bijective

Definition 2.1.4. Let $f : X \xrightarrow{\sim} Y$ be given. The **inverse function** $f^{-1} : Y \to X$ of f is defined as

$$\forall x \in X, f^{-1}(f(x)) = x$$

If

- $f^{-1}(y) = x_1$
- $f^{-1}(y) = x_2$

then

- $f(x_1) = y$
- $f(x_2) = y$

Since f is injective,

$$\begin{gathered} f(x_1) = f(x_2) \\ \Downarrow \\ x_1 = x_2 \end{gathered}$$

Since f is surjective,

$$\forall y \in Y, \exists x \in X, f(x) = y$$

Therefore, f^{-1} is well-defined.

Theorem 2.1.5. *Let $f : X \xrightarrow{\sim} Y$ be given. Then $f^{-1} : Y \to X$ is bijective.*

Proof. Let $x \in X$ and $y_1, y_2 \in Y$ be given.

- **Injectivity:** If
$$f^{-1}(y_1) = f^{-1}(y_2) = x$$
then, by Definition 2.1.4,
  - $f(x) = y_1$
  - $f(x) = y_2$

  Since f is a function,
$$y_1 = y_2$$

- **Surjectivity:** Since f is a function,
$$\begin{gathered} \forall x \in X, \exists y \in Y, f(x) = y \\ \Downarrow \\ f^{-1}(y) = x \\ \Downarrow \\ \forall x \in X, \exists y \in Y, f^{-1}(y) = x \end{gathered}$$

□

Definition 2.1.6. Let $E \subseteq X$, $F \subseteq Y$, and $f : X \to Y$ be given. The following **restriction functions** are defined as

- The **restriction** $f|_E : E \to Y$ of f to E is defined as
$$\forall x \in E, f|_E(x) = f(x)$$

- The **corestriction** $f|^F : X \to F$ of f to F is defined as
$$\forall x \in X, f|^F(x) = f(x)$$
for $f(X) \subseteq F$

- The **bi-restriction** $f|_E^F : E \to F$ of f to E and F is defined as
$$\forall x \in E, f|_E^F(x) = f(x)$$
for $f(E) \subseteq F$

Theorem 2.1.7. *Let $f : X \to Y$ be given. Let $E \subseteq X$ and $F \subseteq Y$ be such that*

$$f(E) \subseteq F$$

Then the following properties hold:

(A) *f is injective $\implies$ $f|_E^F$ is injective*

(B) *$f(E) = F$ $\implies$ $f|_E^F$ is surjective*

(C) *If*

- *f is injective*
- *$f(E) = F$*

then $f|_E^F$ is bijective

Proof. Let $x_1, x_2 \in E$ be given.

(A) If f is injective, then

$$\begin{gathered} f|_E^F(x_1) = f|_E^F(x_2) \\ \Downarrow \\ f(x_1) = f(x_2) \\ \Downarrow \\ x_1 = x_2 \end{gathered}$$

(B) If $f(E) = F$, then

$$\begin{gathered} \forall y \in F, \exists x \in E, f(x) = y \\ \Downarrow \\ \forall y \in F, \exists x \in E, f|_E^F(x) = y \end{gathered}$$

(C) follows from **(A)** and **(B)**

$\square$

Definition 2.1.8. Let $f : X \to Y$ and $g : Y \to Z$ be given. The **composition** $g \circ f : X \to Z$ of g with f is defined as

$$\forall x \in X, (g \circ f)(x) = g(f(x))$$

Theorem 2.1.9. *Let $f : X \to Y$ and $g : Y \to Z$ be given. Then the following properties hold:*

(A) *f and g are injective $\implies$ $g \circ f$ is injective*

(B) *f and g are surjective $\implies$ $g \circ f$ is surjective*

(C) *f and g are bijective $\implies$ $g \circ f$ is bijective*

Proof. Let $x_1, x_2 \in X$ be given.

(A) If f and g are injective, then

$$\begin{gathered}(g \circ f)(x_1) = (g \circ f)(x_2) \\ \Downarrow \\ g(f(x_1)) = g(f(x_2)) \\ \Downarrow \\ f(x_1) = f(x_2) \\ \Downarrow \\ x_1 = x_2\end{gathered}$$

(B) If f and g are surjective, then

$$\forall z \in Z, \exists y \in Y, g(y) = z$$

Also,

$$\forall y \in Y, \exists x \in X, f(x) = y$$

Therefore,

$$\forall z \in Z, \exists x \in X, (g \circ f)(x) = z$$

(C) follows from **(A)** and **(B)**

∎

Theorem 2.1.10. *Let S be a set of sets. Define*

$$\forall X, Y \in S, X \sim Y \iff \exists f : X \xrightarrow{\sim} Y$$

Then $\sim$ is an equivalence relation on S.

Proof. Let $X, Y, Z \in S$ be given.

- **Reflexivity:**

$$\begin{gathered}\forall X \in S, \exists f : X \xrightarrow{\sim} X, f(x) = x \\ \Downarrow \\ \forall X \in S, X \sim X\end{gathered}$$

- **Symmetry:** If $X \sim Y$, by Theorem 2.1.5,

$$\begin{gathered}\exists f : X \xrightarrow{\sim} Y \\ \Downarrow \\ \exists f^{-1} : Y \xrightarrow{\sim} X\end{gathered}$$

Therefore,

$$\forall X, Y \in S, X \sim Y \implies Y \sim X$$

- **Transitivity:** If $X \sim Y$ and $Y \sim Z$, by Theorem 2.1.5,

$$\begin{gathered}\exists f : X \xrightarrow{\sim} Y \wedge \exists g : Y \xrightarrow{\sim} Z \\ \Downarrow \\ \exists g \circ f : X \xrightarrow{\sim} Z\end{gathered}$$

Therefore,

$$\forall X, Y, Z \in S, X \sim Y \wedge Y \sim Z \implies X \sim Z$$

□

2.2 Finite Sets

Recall the relation $\sim$ defined in Theorem 2.1.10. A set X is said to be **finite** if

$$\exists n \in \mathbb{N}_0, \mathbb{N}^n \sim X$$

In this case, n is called the **cardinality** of X, which is denoted by

$$|X| = n$$

A set X is said to be

- **infinite** if it is not finite
- **countable** if $\mathbb{N} \sim X$
- **at most countable** if it is either finite or countable
- **uncountable** if it is neither finite nor countable

Remark. In this textbook, we will use the following notations:

- X is finite $\iff |X| < \aleph_0$
- X is infinite $\iff |X| \geq \aleph_0$
- X is countable $\iff |X| = \aleph_0$
- X is at most countable $\iff |X| \leq \aleph_0$
- X is uncountable $\iff |X| > \aleph_0$

The order symbols above are not **total orders**; they are just convenient notations.

Lemma 2.2.1.

$$\forall n \in \mathbb{N}_0, \forall A \subseteq \mathbb{N}^n, |A| < \aleph_0$$

Proof. Define

$$P(n) : \forall A \subseteq \mathbb{N}^n, |A| < \aleph_0$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}_0$.

- **Base case:** Define $f : \mathbb{N}^0 \xrightarrow{\sim} \varnothing$ as

  $$\forall k \in \mathbb{N}^0, f(k) = k$$

  Then

  $$|\varnothing| = 0 < \aleph_0$$

  Therefore, $P(0)$ is true.

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}_0$. Let $A \subseteq \mathbb{N}^{n+1}$ be given.
  - If $n + 1 \notin A$, then

    $$\begin{gathered} A \subseteq \mathbb{N}^n \\ \Downarrow \\ |A| < \aleph_0 \end{gathered}$$

  - If $n + 1 \in A$, define

    $$B \coloneqq A \setminus \{n + 1\} \subseteq \mathbb{N}^n$$

    Since $B \subseteq \mathbb{N}^n$,

    $$\begin{gathered} B \subseteq \mathbb{N}^n \\ \Downarrow \\ |B| < \aleph_0 \\ \Downarrow \\ \exists m \in \mathbb{N}_0, \exists f : \mathbb{N}^m \xrightarrow{\sim} B \end{gathered}$$

    Define $g : \mathbb{N}^{m+1} \xrightarrow{\sim} A$ as

    $$\forall k \in \mathbb{N}^{m+1}, g(k) = \begin{cases} f(k), & \text{if } k \leq m \\ n + 1, & \text{if } k = m + 1 \end{cases}$$

    Then

    $$|A| = m + 1 < \aleph_0$$

  Therefore, $P(n + 1)$ is true.

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}_0$. □

Theorem 2.2.2.

$$\forall X, \forall Y \subseteq X, |X| < \aleph_0 \implies |Y| < \aleph_0$$

Proof. Let X be finite. Then

$$\begin{gathered} |X| < \aleph_0 \\ \Downarrow \\ \exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n \xrightarrow{\sim} X \end{gathered}$$

Define

$$Z \coloneqq f^{-1}(Y) \subseteq \mathbb{N}^n$$

for $Y \subseteq X$. By Lemma 2.2.1,

$$\begin{gathered} Z \subseteq \mathbb{N}^n \\ \Downarrow \\ |Z| < \aleph_0 \\ \Downarrow \\ \exists m \in \mathbb{N}_0, \exists g : \mathbb{N}^m \xrightarrow{\sim} Z \end{gathered}$$

By Theorems 2.1.7 and 2.1.9,

$$\begin{gathered} f(Z) = Y \\ \Downarrow \\ \exists f|_Z^Y : Z \xrightarrow{\sim} Y \\ \Downarrow \\ \exists f|_Z^Y \circ g : \mathbb{N}^m \xrightarrow{\sim} Y \\ \Downarrow \\ |Y| < \aleph_0 \end{gathered}$$

□

Lemma 2.2.3.

$$\forall Y, (\exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n \twoheadrightarrow Y) \implies |Y| < \aleph_0$$

Proof. Assume that

$$\exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n \twoheadrightarrow Y$$

Define

$$N_y \coloneqq f^{-1}(\{y\}) \subseteq \mathbb{N}^n$$

for each $y \in Y$. By Theorem 1.1.4,

$$\begin{gathered} f \text{ is surjective} \\ \Downarrow \\ \forall y \in Y, N_y \neq \varnothing \\ \Downarrow \\ \exists n_y \in N_y, \forall x \in N_y, n_y \leq x \end{gathered}$$

Define

$$M_Y := \{n_y \mid y \in Y\} \subseteq \mathbb{N}^n$$

Define $g : M_Y \xrightarrow{\sim} Y$ as

$$\forall n_y \in M_Y, g(n_y) = y$$

By Lemma 2.2.1 and theorem 2.1.9,

$$\begin{gathered} |M_Y| < \aleph_0 \\ \Downarrow \\ \exists m \in \mathbb{N}_0, \exists h : \mathbb{N}^m \xrightarrow{\sim} M_Y \\ \Downarrow \\ \exists g \circ h : \mathbb{N}^m \xrightarrow{\sim} Y \\ \Downarrow \\ |Y| < \aleph_0 \end{gathered}$$

□

Theorem 2.2.4.

$$\forall X \forall Y, (|X| < \aleph_0 \wedge \exists f : X \twoheadrightarrow Y) \implies |Y| < \aleph_0$$

Proof. Assume that

$$|X| < \aleph_0 \wedge \exists f : X \twoheadrightarrow Y$$

By Theorem 2.1.9 and lemma 2.2.3,

$$\begin{gathered} |X| < \aleph_0 \\ \Downarrow \\ \exists n \in \mathbb{N}_0, \exists g : \mathbb{N}^n \xrightarrow{\sim} X \\ \Downarrow \\ \exists f \circ g : \mathbb{N}^n \twoheadrightarrow Y \\ \Downarrow \\ |Y| < \aleph_0 \end{gathered}$$

□

Theorem 2.2.5. *Let X satisfy*

- $0 < |X| < \aleph_0$
- *X is ordered*

Then

- $\exists m \in X, \forall x \in X, m \leq x$

- $\exists M \in X, \forall x \in X, x \leq M$

Proof. Since $0 < |X| < \aleph_0$,

$$\begin{gathered} 0 < |X| < \aleph_0 \\ \Downarrow \\ \exists n \in \mathbb{N}, \exists f : \mathbb{N}^n \xrightarrow{\sim} X \end{gathered}$$

The elements of X can be listed as

$$X = \{a_1, a_2, \ldots, a_n\}$$

where

$$\forall k \in \mathbb{N}, a_k = f(k)$$

Define $P(n)$: If

- $|X| = n$
- X is ordered

then

- $\exists m \in X, \forall x \in X, m \leq x$
- $\exists M \in X, \forall x \in X, x \leq M$

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Let $X = \{a_1\}$ be given.
  - $\forall x \in X, a_1 \leq x$
  - $\forall x \in X, x \leq a_1$

  Therefore, $P(1)$ is true.
- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Let
  - $X = \{a_1, a_2, \ldots, a_{n+1}\}$
  - $Y = \{a_1, a_2, \ldots, a_n\} \subseteq X$

  be given. Then
  - $\exists k \in \mathbb{N}^n, \forall y \in Y, a_k \leq y$
  - $\exists K \in \mathbb{N}^n, \forall y \in Y, y \leq a_K$

  Also,
  - $\forall x \in X, \min\{a_k, a_{n+1}\} \leq x$
  - $\forall x \in X, x \leq \max\{a_K, a_{n+1}\}$

Therefore, $P(n+1)$ is true.

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. $\square$

Lemma 2.2.6. *Let $n, m \in \mathbb{N}_0$ be given. Then*

(A) $(\exists f : \mathbb{N}^n \hookrightarrow \mathbb{N}^m) \implies n \leq m$

(B) $(\exists f : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^m) \implies n \geq m$

(C) $\left(\exists f : \mathbb{N}^n \xrightarrow{\sim} \mathbb{N}^m\right) \implies n = m$

(C) *guarantees the uniqueness of the cardinality.*

$$\begin{gathered} n = |X| \wedge m = |X| \\ \Downarrow \\ \mathbb{N}^n \sim X \wedge \mathbb{N}^m \sim X \\ \Downarrow \\ \mathbb{N}^n \sim \mathbb{N}^m \\ \Downarrow \\ n = m \end{gathered}$$

Proof. **(A)** Define

$$P(n) : \forall m \in \mathbb{N}_0, (\exists f_n : \mathbb{N}^n \hookrightarrow \mathbb{N}^m) \implies n \leq m$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}_0$.

- **Base case:**
$$\begin{gathered} \forall m \in \mathbb{N}_0, 0 \leq m \\ \Downarrow \\ P(0) \text{ is true} \end{gathered}$$

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}_0$. If $m = 0$, then
$$\nexists f_{n+1} : \mathbb{N}^{n+1} \hookrightarrow \mathbb{N}^0$$
Otherwise, let $f_{n+1} : \mathbb{N}^{n+1} \hookrightarrow \mathbb{N}^m$ be given with
$$f_{n+1}(n+1) = x \in \mathbb{N}^m$$
Define $g_m : \mathbb{N}^m \xrightarrow{\sim} \mathbb{N}^m$ as
$$\forall y \in \mathbb{N}^m, g_m(y) = \begin{cases} x, & \text{if } y = m \\ m, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Define $f_n : \mathbb{N}^n \to \mathbb{N}^{m-1}$ as

$$f_n \equiv (g_m \circ f_{n+1})|_{\mathbb{N}^n}^{\mathbb{N}^{m-1}}$$

By Theorems 2.1.7 and 2.1.9,

$$\begin{gathered} f_n \text{ is injective} \\ \Downarrow \\ n \leq m-1 \\ \Downarrow \\ n+1 \leq m \\ \Downarrow \\ P(n+1) \text{ is true} \end{gathered}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}_0$.

(B) Define

$$Q(m) : \forall n \in \mathbb{N}_0, (\exists f_m : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^m) \implies n \geq m$$

We will show that $Q(m)$ is true for all $m \in \mathbb{N}_0$.

- **Base case:**

$$\begin{gathered} \forall n \in \mathbb{N}_0, n \geq 0 \\ \Downarrow \\ Q(0) \text{ is true} \end{gathered}$$

- **Inductive step:** Let $Q(m)$ be true for $m \in \mathbb{N}_0$. If $n = 0$, then

$$\nexists f_{m+1} : \mathbb{N}^0 \twoheadrightarrow \mathbb{N}^{m+1}$$

Otherwise, let $f_{m+1} : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^{m+1}$ be given with

$$f_{m+1}(n) = x \in \mathbb{N}^{m+1}$$

Define $g_{m+1} : \mathbb{N}^{m+1} \xrightarrow{\sim} \mathbb{N}^{m+1}$ as

$$\forall y \in \mathbb{N}^{m+1}, g_{m+1}(y) = \begin{cases} x, & \text{if } y = m+1 \\ m+1, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Define

$$K_{m+1} \coloneqq (g_{m+1} \circ f_{m+1})^{-1}(\{m+1\}) \subseteq \mathbb{N}^n$$

Define $f_m : \mathbb{N}^{n-1} \to \mathbb{N}^m$ as

$$\forall y \in \mathbb{N}^{n-1}, f_m(y) = \begin{cases} (g_{m+1} \circ f_{m+1})(y), & \text{if } y \in \mathbb{N}^{n-1} \setminus K_{m+1} \\ m, & \text{if } y \in K_{m+1} \end{cases}$$

By Theorem 2.1.9,

$$\begin{gathered} f_m \text{ is surjective} \\ \Downarrow \\ n-1 \geq m \\ \Downarrow \\ n \geq m+1 \\ \Downarrow \\ Q(m+1) \text{ is true} \end{gathered}$$

By the *principle of mathematical induction*, $Q(m)$ is true for all $m \in \mathbb{N}_0$.

(C) follows from **(A)** and **(B)**.

□

Lemma 2.2.7. *Let $n, m \in \mathbb{N}_0$ be given. Then*

(A) $n \leq m \implies \exists f : \mathbb{N}^n \hookrightarrow \mathbb{N}^m$

(B) $n \geq m \implies \exists f : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^m$

(C) $n = m \implies \exists f : \mathbb{N}^n \xrightarrow{\sim} \mathbb{N}^m$

This lemma is the converse of Lemma 2.2.6.

Proof. **(A)** If $n \leq m$, define $f : \mathbb{N}^n \hookrightarrow \mathbb{N}^m$ as

$$\forall i \in \mathbb{N}^n, f(i) = i$$

(B) If $n \geq m$, define $f : \mathbb{N}^n \to \mathbb{N}^m$ as

$$\forall i \in \mathbb{N}^n, f(i) = \begin{cases} i, & \text{if } i \leq m \\ m, & \text{if } i > m \end{cases}$$

(C) follows from **(A)** and **(B)**.

□

Theorem 2.2.8. *Let X, Y be finite. Then*

(A) $|X| \leq |Y| \iff \exists f : X \hookrightarrow Y$

(B) $|X| \geq |Y| \iff \exists f : X \twoheadrightarrow Y$

(C) $|X| = |Y| \iff \exists f : X \xrightarrow{\sim} Y$

*The contraposition of **(A,** $\Longleftarrow$ **)** is called the **pigeonhole principle**.*

Proof. Since X and Y are finite,

- $\exists n \in \mathbb{N}_0, \exists f_X : \mathbb{N}^n \xrightarrow{\sim} X$
- $\exists m \in \mathbb{N}_0, \exists f_Y : \mathbb{N}^m \xrightarrow{\sim} Y$

($\Longleftarrow$) Define $f : X \to Y$ and $F : \mathbb{N}^n \to \mathbb{N}^m$ as

$$F \equiv (f_Y)^{-1} \circ f \circ f_X$$

By Theorems 2.1.5 and 2.1.9,

- f is injective $\implies$ F is injective
- f is surjective $\implies$ F is surjective

By Lemma 2.2.6,

(A) $(\exists f : X \hookrightarrow Y) \implies (\exists F : \mathbb{N}^n \hookrightarrow \mathbb{N}^m) \implies n \leq m$

(B) $(\exists f : X \twoheadrightarrow Y) \implies (\exists F : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^m) \implies n \geq m$

(C) follows from **(A)** and **(B)**.

($\implies$) Define $f : X \to Y$ and $G : \mathbb{N}^m \to \mathbb{N}^n$ as

$$f \equiv f_Y \circ G \circ (f_X)^{-1}$$

By Theorems 2.1.5 and 2.1.9,

(A) G is injective $\implies$ f is injective

(B) G is surjective $\implies$ f is surjective

By Lemma 2.2.7,

(A) $n \leq m \implies (\exists G : \mathbb{N}^n \hookrightarrow \mathbb{N}^m) \implies (\exists f : X \hookrightarrow Y)$

(B) $n \geq m \implies (\exists G : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^m) \implies (\exists f : X \twoheadrightarrow Y)$

(C) follows from **(A)** and **(B)**.

□

Lemma 2.2.9.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \wedge X \cap Y = \varnothing \implies |X \cup Y| < \aleph_0$$

Proof. Assume that

$$|X| < \aleph_0 \wedge |Y| < \aleph_0 \wedge X \cap Y = \varnothing$$

Then

- $\exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n \xrightarrow{\sim} X$
- $\exists m \in \mathbb{N}_0, \exists g : \mathbb{N}^m \xrightarrow{\sim} Y$

Define $h : \mathbb{N}^{m+n} \xrightarrow{\sim} X \cup Y$ as

$$\forall i \in \mathbb{N}^{m+n}, h(i) = \begin{cases} f(i), & \text{if } i \leq n \\ g(i-n), & \text{if } n < i \leq n+m \end{cases}$$

Then

$$|X \cup Y| = m + n < \aleph_0$$

□

Theorem 2.2.10.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \cup Y| < \aleph_0$$

Proof. Assume that

$$|X| < \aleph_0 \wedge |Y| < \aleph_0$$

Define

- $A = X \setminus Y \subseteq X$
- $B = Y \setminus X \subseteq Y$
- $C = X \cap Y \subseteq X$

By Theorem 2.2.2,

- $|A| < \aleph_0$, $|B| < \aleph_0$, and $|C| < \aleph_0$
- $A \cap B = \varnothing$, $B \cap C = \varnothing$, and $C \cap A = \varnothing$

Consider the following:

- $X = (X \setminus Y) \cup (X \cap Y) = A \cup C$
- $Y = (Y \setminus X) \cup (X \cap Y) = B \cup C$
- $X \cup Y = ((X \setminus Y) \cup (Y \setminus X)) \cup (X \cap Y) = (A \cup B) \cup C$

By Lemma 2.2.9,

$$\begin{gathered} |A \cup B| < \aleph_0 \\ \Downarrow \\ |(A \cup B) \cup C| < \aleph_0 \\ \Downarrow \\ |X \cup Y| < \aleph_0 \end{gathered}$$

□

Theorem 2.2.11.

$$\forall n \in \mathbb{N}, (\forall k \in \mathbb{N}^n, |X_k| < \aleph_0) \implies \left|\bigcup_{k=1}^{n} X_k\right| < \aleph_0$$

Proof. Define

$$P(n) : (\forall k \in \mathbb{N}^n, |X_k| < \aleph_0) \implies \left|\bigcup_{k=1}^{n} X_k\right| < \aleph_0$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Let $|X_1| < \aleph_0$ be given. Then

$$\begin{gathered} |X_1| < \aleph_0 \\ \Downarrow \\ \left|\bigcup_{k=1}^{1} X_k\right| < \aleph_0 \\ \Downarrow \\ P(1) \text{ is true} \end{gathered}$$

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Assume that

$$\forall k \in \mathbb{N}^{n+1}, |X_k| < \aleph_0$$

  By the inductive hypothesis,

  – $\left|\bigcup_{k=1}^{n} X_k\right| < \aleph_0$
  – $|X_{n+1}| < \aleph_0$

  By Theorem 2.2.10,

$$\begin{gathered} \bigcup_{k=1}^{n+1} X_k = \left(\bigcup_{k=1}^{n} X_k\right) \cup X_{n+1} \\ \Downarrow \\ \left|\bigcup_{k=1}^{n+1} X_k\right| < \aleph_0 \\ \Downarrow \\ P(n+1) \text{ is true} \end{gathered}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. □

Lemma 2.2.12.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \times Y| < \aleph_0$$

Proof. Assume that

$$|X| < \aleph_0 \wedge |Y| < \aleph_0$$

Then

- $\exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n \xrightarrow{\sim} X$
- $\exists m \in \mathbb{N}_0, \exists g : \mathbb{N}^m \xrightarrow{\sim} Y$

Define

$$i = (a-1)m + b \in \mathbb{N}^{nm}$$

for each $a \in \mathbb{N}^n$ and $b \in \mathbb{N}^m$. Define $h : \mathbb{N}^{nm} \xrightarrow{\sim} X \times Y$ as

$$\forall i \in \mathbb{N}^{nm}, h(i) = (f(a), g(b))$$

Then

$$|X \times Y| = nm < \aleph_0$$

□

Theorem 2.2.13.

$$\forall n \in \mathbb{N}, (\forall k \in \mathbb{N}^n, |X_k| < \aleph_0) \implies \left|\prod_{k=1}^{n} X_k\right| < \aleph_0$$

Proof. Define

$$P(n) : (\forall k \in \mathbb{N}^n, |X_k| < \aleph_0) \implies \left|\prod_{k=1}^{n} X_k\right| < \aleph_0$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Let $|X_1| < \aleph_0$ be given. Then

$$\begin{gathered} |X_1| < \aleph_0 \\ \Downarrow \\ \left|\prod_{k=1}^{1} X_k\right| < \aleph_0 \\ \Downarrow \\ P(1) \text{ is true} \end{gathered}$$

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Assume that

$$\forall k \in \mathbb{N}^{n+1}, |X_k| < \aleph_0$$

By the inductive hypothesis,

  - $\left|\prod_{k=1}^{n} X_k\right| < \aleph_0$
  - $|X_{n+1}| < \aleph_0$

By Lemma 2.2.12,

$$\begin{gathered} \prod_{k=1}^{n+1} X_k \cong \left(\prod_{k=1}^{n} X_k\right) \times X_{n+1} \\ \Downarrow \\ \left|\prod_{k=1}^{n+1} X_k\right| < \aleph_0 \\ \Downarrow \\ P(n+1) \text{ is true} \end{gathered}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. □

Theorem 2.2.14.

$$\forall X, \nexists f : X \twoheadrightarrow \mathcal{P}(X)$$

where

$$\mathcal{P}(X) \coloneqq \{E \mid E \subseteq X\}$$

is called the ***power set*** *of X. This theorem is called* ***Cantor's theorem***.

Proof. Assume that

$$\exists f : X \twoheadrightarrow \mathcal{P}(X)$$

Define

$$S \coloneqq \{x \in X \mid x \notin f(x)\} \subseteq X$$

Then

$$\begin{gathered} S \subseteq X \\ \Downarrow \\ S \in \mathcal{P}(X) \\ \Downarrow \\ \exists x \in X, f(x) = S \end{gathered}$$

There are two cases:

- If $x \in S$, then

$$\begin{gathered} x \in S \\ \Downarrow \\ x \notin f(x) = S \\ \Downarrow \\ \perp \end{gathered}$$

- If $x \notin S$, then

$$\begin{gathered} x \notin S \\ \Downarrow \\ x \in f(x) = S \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\nexists f : X \twoheadrightarrow \mathcal{P}(X)$$

□

2.3 Countable and Uncountable Sets

Theorem 2.3.1.

$$\forall X \forall Y, |X| \geq \aleph_0 \wedge |Y| < \aleph_0 \implies |X \setminus Y| \geq \aleph_0$$

Proof. Assume that

$$\exists X \exists Y, |X| \geq \aleph_0 \wedge |Y| < \aleph_0 \wedge |X \setminus Y| < \aleph_0$$

By Theorem 2.2.10,

$$|(X \setminus Y) \cup Y| < \aleph_0$$

By Theorem 2.2.2,

$$\begin{gathered} X = (X \setminus Y) \cup (X \cap Y) \\ \Downarrow \\ X \subseteq (X \setminus Y) \cup Y \\ \Downarrow \\ |X| < \aleph_0 \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\forall X \forall Y, |X| \geq \aleph_0 \wedge |Y| < \aleph_0 \implies |X \setminus Y| \geq \aleph_0$$

□

Theorem 2.3.2.

$$\forall X \forall Y, |X| \geq \aleph_0 \land (\exists f : X \hookrightarrow Y) \implies |Y| \geq \aleph_0$$

Proof. Assume that

$$\exists X \exists Y, |X| \geq \aleph_0 \land (\exists f : X \hookrightarrow Y) \land |Y| < \aleph_0$$

Define $g : Y \twoheadrightarrow X$ as

$$\forall y \in Y, g(y) = \begin{cases} f^{-1}(y), & \text{if } y \in f(X) \\ f^{-1}(y_0), & \text{otherwise} \end{cases}$$

where $y_0 \in f(X)$ is fixed. By Theorem 2.2.4,

$$\begin{gathered} \exists g : Y \twoheadrightarrow X \\ \Downarrow \\ |X| < \aleph_0 \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\forall X \forall Y, |X| \geq \aleph_0 \land (\exists f : X \hookrightarrow Y) \implies |Y| \geq \aleph_0$$

□

Lemma 2.3.3.

$$|\mathbb{N}| \geq \aleph_0$$

Proof. Assume that

$$|\mathbb{N}| < \aleph_0$$

By Theorem 2.2.5,

$$\exists m \in \mathbb{N}, \forall x \in \mathbb{N}, x \leq m$$

Then

$$\begin{gathered} m < m + 1 \\ \Downarrow \\ m + 1 \notin \mathbb{N} \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$|\mathbb{N}| \geq \aleph_0$$

□

Theorem 2.3.4.

$$\forall X, |X| = \aleph_0 \implies |X| \geq \aleph_0$$

Proof. Let $|X| = \aleph_0$ be given. Then

$$\exists f : \mathbb{N} \xrightarrow{\sim} X$$

By Theorem 2.3.2 and lemma 2.3.3,

$$|X| \geq \aleph_0$$

□

Theorem 2.3.5.

$$|\mathbb{Z}| = \aleph_0$$

Proof. Define $f : \mathbb{N} \to \mathbb{Z}$ as

$$\forall n \in \mathbb{N}, f(n) = \begin{cases} -\dfrac{n}{2}, & \text{if } n \text{ is even} \\ \dfrac{n-1}{2}, & \text{if } n \text{ is odd} \end{cases}$$

We will show that

$$f \text{ is bijective} \\ \Downarrow \\ |\mathbb{Z}| = \aleph_0$$

- **Injectivity:** Let $f(n_1) = f(n_2)$ be given for $n_1, n_2 \in \mathbb{N}$. Since
  - n is even $\implies f(n) = -\dfrac{n}{2} < 0$
  - n is odd $\implies f(n) = \dfrac{n-1}{2} \geq 0$

  n_1 and n_2 are both even or both odd.
  - If both n_1 and n_2 are even, then

  $$-\frac{n_1}{2} = -\frac{n_2}{2} \\ \Downarrow \\ n_1 = n_2$$

  - If both n_1 and n_2 are odd, then

  $$\frac{n_1 - 1}{2} = \frac{n_2 - 1}{2} \\ \Downarrow \\ n_1 = n_2$$

- **Surjectivity:** Let $k \in \mathbb{Z}$ be given.
  - If $k \geq 0$, then

$$\begin{gathered} 2k+1 \in \mathbb{N} \\ \Downarrow \\ f(2k+1) = \frac{(2k+1)-1}{2} = \frac{2k}{2} = k \end{gathered}$$

  - If $k < 0$, then

$$\begin{gathered} -2k \in \mathbb{N} \\ \Downarrow \\ f(-2k) = -\frac{-2k}{2} = k \end{gathered}$$

$\square$

Lemma 2.3.6.

$$\forall X \subseteq \mathbb{N}, |X| \geq \aleph_0 \implies |X| = \aleph_0$$

Proof. Assume that

- $X \subseteq \mathbb{N}$
- $|X| \geq \aleph_0$

Define $(X_n)_{n=1}^{\infty}$ and $f : \mathbb{N} \to X$ as

$$\begin{gathered} X_1 = X \subseteq \mathbb{N} \\ X_2 = X_1 \setminus \{f(1)\} \subseteq \mathbb{N} \\ \vdots \\ X_{n+1} = X_n \setminus \{f(n)\} \subseteq \mathbb{N} \end{gathered} \qquad \begin{gathered} f(1) = \min X_1 \\ f(2) = \min X_2 \\ \vdots \\ f(n+1) = \min X_{n+1} \end{gathered}$$

for each $n \in \mathbb{N}$. By Theorems 1.1.4 and 2.3.1,

$$\begin{gathered} \forall n \in \mathbb{N}, |X_n| \geq \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, X_n \neq \varnothing \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists f(n) = \min X_n \in X_n \\ \Downarrow \\ f \text{ is well-defined} \end{gathered}$$

We will show that

$$\begin{gathered} f \text{ is bijective} \\ \Downarrow \\ |X| = \aleph_0 \end{gathered}$$

- **Injectivity:** Let $n_1 \neq n_2$ (*WLOG* $n_1 < n_2$) be given for $n_1, n_2 \in \mathbb{N}$.
  - $f(n_1) = \min X_{n_1} \in X_{n_1}$
  - $f(n_2) = \min X_{n_2} \in X_{n_2}$

  Then
  $$\begin{gathered} X_{n_2} \subseteq X_{n_1} \setminus \{f(n_1)\} \\ \Downarrow \\ f(n_2) \neq f(n_1) \\ \Downarrow \\ f \text{ is injective} \end{gathered}$$

- **Surjectivity:** Let $x \in X$ be given. Define
  $$S \coloneqq \{n \in X \mid n < x\}$$
  By Lemma 2.2.1,
  $$\begin{gathered} S \subseteq \mathbb{N}^{x-1} \\ \Downarrow \\ |S| < \aleph_0 \\ \Downarrow \\ \exists k \in \mathbb{N}_0, |S| = k \end{gathered}$$
  If $k = 0$, then
  - $S = \varnothing$
  - $x = \min X_1 = f(1)$

  Otherwise,
  - $S = \{s_1, s_2, \ldots, s_k\}$
  - $s_1 < s_2 < \cdots < s_k$

  Then
  $$\begin{gathered} \min X_1 = \min S \\ \min X_2 = \min S \setminus \{s_1\} \\ \vdots \\ \min X_k = \min S \setminus \{s_1, s_2, \ldots, s_{k-1}\} \end{gathered} \qquad \begin{gathered} X_2 = X \setminus \{s_1\} \\ X_3 = X \setminus \{s_1, s_2\} \\ \vdots \\ X_{k+1} = X \setminus S \end{gathered}$$
  Therefore,
  $$\begin{gathered} f(k+1) = \min X_{k+1} = x \\ \Downarrow \\ f \text{ is surjective} \end{gathered}$$

$\square$

Theorem 2.3.7.

$$\forall X, \forall Y \subseteq X, |X| = \aleph_0 \land |Y| \geq \aleph_0 \implies |Y| = \aleph_0$$

Proof. Assume that

- $Y \subseteq X$
- $|X| = \aleph_0$
- $|Y| \geq \aleph_0$

By Theorem 2.1.7,

$$\begin{gathered} \exists f : \mathbb{N} \xrightarrow{\sim} X \\ \Downarrow \\ \exists f|_{f^{-1}(Y)}^{Y} : f^{-1}(Y) \xrightarrow{\sim} Y \end{gathered}$$

By Theorem 2.2.4, if $f^{-1}(Y)$ is finite, then

$$\begin{gathered} \left|f^{-1}(Y)\right| < \aleph_0 \\ \Downarrow \\ |Y| < \aleph_0 \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\left|f^{-1}(Y)\right| \geq \aleph_0$$

By Lemma 2.3.6,

$$\begin{gathered} \left|f^{-1}(Y)\right| \subseteq \mathbb{N} \\ \Downarrow \\ \left|f^{-1}(Y)\right| = \aleph_0 \\ \Downarrow \\ \exists g : \mathbb{N} \xrightarrow{\sim} f^{-1}(Y) \\ \Downarrow \\ \exists f|_{f^{-1}(Y)}^{Y} \circ g : \mathbb{N} \xrightarrow{\sim} Y \\ \Downarrow \\ |Y| = \aleph_0 \end{gathered}$$

□

Theorem 2.3.8.

$$\forall X, |X| \geq \aleph_0 \land (\exists f : \mathbb{N} \twoheadrightarrow X) \implies |X| = \aleph_0$$

Proof. Assume that

- $|X| \geq \aleph_0$
- $\exists f : \mathbb{N} \twoheadrightarrow X$

Define

$$S_k \coloneqq f^{-1}(\{k\})$$

for each $k \in X$. By Theorem 1.1.4,

$$\begin{gathered} f \text{ is surjective} \\ \Downarrow \\ \forall k \in X, S_k \neq \varnothing \\ \Downarrow \\ \forall k \in X, \exists m \in S_k, m = \min S_k \end{gathered}$$

Define $g : X \to \mathbb{N}$ as

$$\forall k \in X, g(k) = \min S_k$$

Let $k_1, k_2 \in X$ be given. Then

$$\begin{gathered} \forall i, j \in X, i \neq j \implies S_i \cap S_j = \varnothing \\ \Downarrow \\ g(k_1) = g(k_2) \implies \min S_{k_1} = \min S_{k_2} \implies k_1 = k_2 \\ \Downarrow \\ g \text{ is injective} \end{gathered}$$

By Theorem 2.1.7,

$$\exists g|^{g(X)} : X \xrightarrow{\sim} g(X)$$

By Theorem 2.3.2,

$$\begin{gathered} |X| \geq \aleph_0 \\ \Downarrow \\ |g(X)| \geq \aleph_0 \end{gathered}$$

By Lemma 2.3.6,

$$\begin{gathered} g(X) \subseteq \mathbb{N} \\ \Downarrow \\ |g(X)| = \aleph_0 \\ \Downarrow \\ |X| = \aleph_0 \end{gathered}$$

□

Theorem 2.3.9.

$$(\forall n \in \mathbb{N}, |X_n| < \aleph_0) \implies \left| \bigcup_{n=1}^{\infty} X_n \right| \leq \aleph_0$$

Proof. Assume that

$$\begin{gathered}\forall n \in \mathbb{N}, |X_n| < \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists m_n \in \mathbb{N}_0, \exists g_n : \mathbb{N}^{m_n} \xrightarrow{\sim} X_n\end{gathered}$$

If

$$\left|\bigcup_{n=1}^{\infty} X_n\right| < \aleph_0$$

then the proof is complete. Otherwise,

$$\left|\bigcup_{n=1}^{\infty} X_n\right| \geq \aleph_0$$

Define $h : \mathbb{N} \twoheadrightarrow \bigcup\limits_{n=1}^{\infty} X_n$ as

- $\forall m \in \mathbb{N}_{m_1}, h(m) = g_1(m)$
- $\forall n \in \mathbb{N}_2, \forall m \in \mathbb{N}_{m_n}, h\left(\sum\limits_{k=1}^{n-1} m_i + m\right) = g_n(m)$

By Theorem 2.3.8,

$$\begin{gathered}\left|\bigcup_{n=1}^{\infty} X_n\right| \geq \aleph_0 \\ \Downarrow \\ \left|\bigcup_{n=1}^{\infty} X_n\right| = \aleph_0\end{gathered}$$

□

Theorem 2.3.10.

$$(\forall n \in \mathbb{N}, |X_n| = \aleph_0) \implies \left|\bigcup_{n=1}^{\infty} X_n\right| = \aleph_0$$

Proof. Assume that

$$\left|\bigcup_{n=1}^{\infty} X_n\right| < \aleph_0$$

By Theorem 2.2.2,

$$\begin{gathered} X_1 \subseteq \bigcup_{n=1}^{\infty} X_n \\ \Downarrow \\ |X_1| < \aleph_0 \\ \Downarrow \\ \bot \end{gathered}$$

Therefore,

$$\left| \bigcup_{n=1}^{\infty} X_n \right| \geq \aleph_0$$

Define

$$T_n := \left\{ k \in \mathbb{N} \,\middle|\, n \leq \frac{k(k+1)}{2} \right\}$$

for each $n \in \mathbb{N}$. By Theorem 1.1.4,

$$\begin{gathered} 2n \in T_n \\ \Downarrow \\ T_n \neq \varnothing \\ \Downarrow \\ \exists t_n \in T_n, t_n = \min T_n \end{gathered}$$

for all $n \in \mathbb{N}$. Define

$$s_n := n - \frac{t_n(t_n - 1)}{2} = n - \frac{t_n(t_n + 1)}{2} + t_n \in \mathbb{N}$$

for each $n \in \mathbb{N}$. Then

$$\begin{gathered} \frac{t_n(t_n - 1)}{2} < n \leq \frac{t_n(t_n + 1)}{2} \\ \Downarrow \\ t_n - s_n + 1 = \frac{t_n(t_n + 1)}{2} - n + 1 \geq 1 \\ \Downarrow \\ t_n - s_n + 1 \in \mathbb{N} \end{gathered}$$

for all $n \in \mathbb{N}$. Assume that

$$\begin{gathered} \forall n \in \mathbb{N}, |X_n| = \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists f_n : \mathbb{N} \xrightarrow{\sim} X_n \end{gathered}$$

Define $g : \mathbb{N} \to \bigcup_{n=1}^{\infty} X_n$ as

$$\forall n \in \mathbb{N}, g(n) = f_{s_n}(t_n - s_n + 1)$$

Let $x \in \bigcup_{n=1}^{\infty} X_n$ be given. Then

$$\begin{gathered}\exists m \in \mathbb{N}, x \in X_m \\ \Downarrow \\ \exists l \in \mathbb{N}, f_m(l) = x\end{gathered}$$

Define

$$k := \frac{(l+m-1)(l+m-2)}{2} + m \in \mathbb{N}$$

We will show that

$$\begin{gathered}l+m-2 \notin T_k \wedge l+m-1 \in T_k \\ \Downarrow \\ t_k = \min T_k = l+m-1 \\ \Downarrow \\ s_k = k - \frac{t_k(t_k-1)}{2} = m \\ \Downarrow \\ g(k) = f_{s_k}(t_k - s_k + 1) = f_m(l+m-1-m+1) = f_m(l) = x\end{gathered}$$

Consider the following inequalities:

- Let $z = l+m-2$ be given. Then

$$\begin{aligned}k - \frac{z(z+1)}{2} &= k - \frac{(l+m-2)(l+m-1)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-2)(l+m-1)}{2} \\ &= m > 0\end{aligned}$$

  Therefore,

$$z = l+m-2 \notin T_k$$

- Let $z = l+m-1$ be given. Then

$$\begin{aligned}k - \frac{z(z+1)}{2} &= k - \frac{(l+m-1)(l+m)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-1)(l+m)}{2} \\ &= m - \frac{(l+m-1)(l+m) - (l+m-1)(l+m-2)}{2} \\ &= m - (l+m-1) = 1 - l \leq 0\end{aligned}$$

  Therefore,

$$z = l+m-1 \in T_k$$

By Theorem 2.3.8,

$$\begin{gathered} g \text{ is surjective} \\ \Downarrow \\ \left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0 \end{gathered}$$

□

Corollary 2.3.11.

$$\forall n \in \mathbb{N}, (\forall k \in \mathbb{N}^n, |X_k| = \aleph_0) \implies \left| \bigcup_{k=1}^{n} X_k \right| = \aleph_0$$

Proof. Assume that

$$(\forall k \in \mathbb{N}^n, |X_k| = \aleph_0)$$

for $n \in \mathbb{N}$. Define

$$X_k \coloneqq X_n$$

for all $k \in \mathbb{N}_{n+1}$. By Theorem 2.3.10,

$$\begin{gathered} \forall n \in \mathbb{N}, |X_n| = \aleph_0 \\ \Downarrow \\ \left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0 \end{gathered}$$

Therefore,

$$\begin{gathered} \bigcup_{k=1}^{n} X_k = \bigcup_{n=1}^{\infty} X_n \\ \Downarrow \\ \left| \bigcup_{k=1}^{n} X_k \right| = \aleph_0 \end{gathered}$$

□

Lemma 2.3.12.

$$|\mathbb{N} \times \mathbb{N}| = \aleph_0$$

Proof. Define $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ as

$$\forall m, n \in \mathbb{N}, f(m, n) = \frac{(m+n-1)(m+n-2)}{2} + m$$

We will show that f is bijective.

- **Injectivity:** Let $(m_1, n_1) \neq (m_2, n_2)$ be given for $m_1, n_1, m_2, n_2 \in \mathbb{N}$. If
$$m_1 + n_1 = m_2 + n_2$$
then
$$\begin{gathered} m_1 \neq m_2 \\ \Downarrow \\ f(m_1, n_1) \neq f(m_2, n_2) \end{gathered}$$
Otherwise, $m_1 + n_1 \neq m_2 + n_2$ (*WLOG* $m_1 + n_1 < m_2 + n_2$). Then
$$\begin{aligned} f(m_1, n_1) &= \frac{(m_1+n_1-1)(m_1+n_1-2)}{2} + m_1 \\ &\leq \frac{(m_1+n_1-1)(m_1+n_1-2)}{2} + (m_1+n_1-1) \\ &= \frac{(m_1+n_1)(m_1+n_1-1)}{2} \\ &\leq \frac{(m_2+n_2-1)(m_2+n_2-2)}{2} \\ &< \frac{(m_2+n_2-1)(m_2+n_2-2)}{2} + m_2 = f(m_2, n_2) \end{aligned}$$

- **Surjectivity:** Let $k \in \mathbb{N}$ be given. Define t_k, s_k as in the proof of Theorem 2.3.10. Then
$$f(s_k, t_k - s_k + 1) = k$$

Therefore,
$$|\mathbb{N} \times \mathbb{N}| = \aleph_0$$
□

Theorem 2.3.13.
$$\forall X \forall Y, |X| = \aleph_0 \land |Y| = \aleph_0 \implies |X \times Y| = \aleph_0$$

Proof. Since X and Y are countable,

- $\exists f : \mathbb{N} \xrightarrow{\sim} X$
- $\exists g : \mathbb{N} \xrightarrow{\sim} Y$

Define $h : \mathbb{N} \times \mathbb{N} \xrightarrow{\sim} X \times Y$ as
$$\forall m, n \in \mathbb{N}, h(m, n) = (f(m), g(n))$$
By Lemma 2.3.12,
$$|X \times Y| = \aleph_0$$
□

Theorem 2.3.14.

$$\forall n \in \mathbb{N}, (\forall k \in \mathbb{N}^n, |X_k| = \aleph_0) \implies \left| \prod_{k=1}^{n} X_k \right| = \aleph_0$$

Proof. Define

$$P(n) : (\forall k \in \mathbb{N}^n, |X_k| = \aleph_0) \implies \left| \prod_{k=1}^{n} X_k \right| = \aleph_0$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Let $|X_1| = \aleph_0$ be given. Then

$$\begin{gathered} |X_1| = \aleph_0 \\ \Downarrow \\ \left| \prod_{k=1}^{1} X_k \right| = \aleph_0 \\ \Downarrow \\ P(1) \text{ is true} \end{gathered}$$

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Assume that

$$\forall k \in \mathbb{N}^{n+1}, |X_k| = \aleph_0$$

  By the inductive hypothesis,

  - $\left| \prod_{k=1}^{n} X_k \right| = \aleph_0$
  - $|X_{n+1}| = \aleph_0$

  By Theorem 2.3.13,

$$\begin{gathered} \prod_{k=1}^{n+1} X_k \cong \left(\prod_{k=1}^{n} X_k \right) \times X_{n+1} \\ \Downarrow \\ \left| \prod_{k=1}^{n+1} X_k \right| = \aleph_0 \\ \Downarrow \\ P(n+1) \text{ is true} \end{gathered}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. $\square$

Theorem 2.3.15.
$$|\mathbb{Q}| = \aleph_0$$

Proof. Let $q \in \mathbb{Q}$ be given. Then

$$\exists m \in \mathbb{Z}, \exists n \in \mathbb{N}, q = \frac{m}{n}$$

Define $f : \mathbb{Z} \times \mathbb{N} \twoheadrightarrow \mathbb{Q}$ as

$$\forall (m, n) \in \mathbb{Z} \times \mathbb{N}, f(m, n) = \frac{m}{n}$$

By Theorems 2.3.5 and 2.3.13,

$$\begin{gathered} |\mathbb{Z} \times \mathbb{N}| = \aleph_0 \\ \Downarrow \\ \exists g : \mathbb{N} \xrightarrow{\sim} \mathbb{Z} \times \mathbb{N} \end{gathered}$$

By Lemma 2.3.3 and theorem 2.2.2,

$$\begin{gathered} \mathbb{N} \subseteq \mathbb{Q} \\ \Downarrow \\ |\mathbb{Q}| \geq \aleph_0 \end{gathered}$$

By Theorems 2.1.9 and 2.3.8,

$$\begin{gathered} \exists f \circ g : \mathbb{N} \twoheadrightarrow \mathbb{Q} \\ \Downarrow \\ |\mathbb{Q}| = \aleph_0 \end{gathered}$$

□

Lemma 2.3.16.
$$\left| \prod_{n=1}^{\infty} \{0, 1\} \right| > \aleph_0$$

Proof. Define $(e_n)_{n=1}^{\infty}$ in $\displaystyle\prod_{n=1}^{\infty} \{0, 1\}$ as

$$\forall m \in \mathbb{N}, e_{nm} = \begin{cases} 1, & \text{if } m = n \\ 0, & \text{otherwise} \end{cases}$$

for each $n \in \mathbb{N}$. Define $f : \mathbb{N} \hookrightarrow \displaystyle\prod_{n=1}^{\infty} \{0, 1\}$ as

$$\forall n \in \mathbb{N}, f(n) = e_n$$

By Lemma 2.3.3 and theorem 2.3.2,

$$\left|\prod_{n=1}^{\infty}\{0,1\}\right| \geq \aleph_0$$

If $\displaystyle\prod_{n=1}^{\infty}\{0,1\}$ is countable, then

$$\exists g : \mathbb{N} \xrightarrow{\sim} \prod_{n=1}^{\infty}\{0,1\}$$

Let

- $\displaystyle\prod_{n=1}^{\infty}\{0,1\} = \{a_1, a_2, a_3, \dots\}$
- $a_1 = (a_{11}, a_{12}, a_{13}, \dots)$
- $a_2 = (a_{21}, a_{22}, a_{23}, \dots)$

  $\vdots$

be given, where

$$\forall n \in \mathbb{N}, a_n = g(n)$$

Define $b = (b_1, b_2, b_3, \dots) \in \displaystyle\prod_{n=1}^{\infty}\{0,1\}$ as

$$\forall n \in \mathbb{N}, b_n = 1 - a_{nn}$$

Then

$$\begin{gathered}\forall n \in \mathbb{N}, b \neq a_n \\ \Downarrow \\ b \notin \{a_1, a_2, a_3, \dots\} = \prod_{n=1}^{\infty}\{0,1\} \\ \Downarrow \\ g \text{ is not surjective} \\ \Downarrow \\ \perp\end{gathered}$$

Therefore,

$$\left|\prod_{n=1}^{\infty}\{0,1\}\right| > \aleph_0$$

$\square$

Theorem 2.3.17.

$$|\mathbb{R}| > \aleph_0$$

Proof. By Lemma 2.3.3 and theorem 2.2.2,

$$\begin{gathered}\mathbb{N} \subseteq \mathbb{R} \\ \Downarrow \\ |\mathbb{R}| \geq \aleph_0\end{gathered}$$

Define $f : \mathbb{N} \hookrightarrow [0, 1)$ as

$$\forall n \in \mathbb{N}, f(n) = \frac{1}{n+1}$$

By Lemma 2.3.3 and theorem 2.3.2,

$$|[0, 1)| \geq \aleph_0$$

By Theorem 2.3.7,

$$\begin{gathered}[0, 1) \subseteq \mathbb{R} \\ \Downarrow \\ |[0, 1)| > \aleph_0 \implies |\mathbb{R}| > \aleph_0\end{gathered}$$

Define $g : [0, 1) \to \prod_{n=1}^{\infty} \{0, 1\}$ as

$$\forall x \in [0, 1), g(x) = (a_n^x)_{n=1}^{\infty}$$

where

$$(a_n)_{n=1}^{\infty} \text{ is the } \textit{greedy base } 2 \textit{ expansion} \text{ of } x$$

Define

$$Y := g([0, 1)) \subseteq \prod_{n=1}^{\infty} \{0, 1\}$$

By Theorems 1.6.9 and 2.1.7,

$$\exists g|^Y : [0, 1) \xrightarrow{\sim} Y$$

If $[0, 1)$ is countable, then

$$\exists h : \mathbb{N} \xrightarrow{\sim} [0, 1)$$

By Theorem 2.1.9,

$$\begin{gathered}g|^Y \circ h : \mathbb{N} \xrightarrow{\sim} Y \\ \Downarrow \\ |Y| = \aleph_0\end{gathered}$$

Define

$$Z := \prod_{n=1}^{\infty} \{0, 1\} \setminus Y$$

By Theorems 1.6.10 and 1.6.11,

$$Z = \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0,1\} \;\middle|\; \exists N \in \mathbb{N}, \forall n \in \mathbb{N}_N, a_n = 1 \right\}$$

Define

$$Z_N := \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0,1\} \;\middle|\; \forall n \in \mathbb{N}_N, a_n = 1 \right\} \neq \varnothing$$

for each $N \in \mathbb{N}$. Define $F_N : \prod_{n=1}^{N-1} \{0,1\} \twoheadrightarrow Z_N$ as

$$\forall (a_n)_{n=1}^{N-1} \in \prod_{n=1}^{N-1} \{0,1\}, F_N\left((a_n)_{n=1}^{N-1}\right) = (a_n)_{n=1}^{N-1} + (1)_{n=N}^{\infty}$$

By Theorems 2.2.4 and 2.2.13,

$$\forall N \in \mathbb{N}, |Z_N| < \aleph_0$$

By definition,

$$\begin{gathered} \forall N, M \in \mathbb{N}, N \neq M \implies Z_N \cap Z_M = \varnothing \\ \Downarrow \\ \left| \bigcup_{N=1}^{\infty} Z_N \right| \geq \aleph_0 \end{gathered}$$

By Theorem 2.3.9,

$$\begin{gathered} Z = \bigcup_{N=1}^{\infty} Z_N \\ \Downarrow \\ |Z| = \aleph_0 \end{gathered}$$

By Corollary 2.3.11 and lemma 2.3.16,

$$\begin{gathered} \prod_{n=1}^{\infty} \{0,1\} = Y \cup Z \\ \Downarrow \\ \left| \prod_{n=1}^{\infty} \{0,1\} \right| = \aleph_0 \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\begin{gathered} |[0,1)| > \aleph_0 \\ \Downarrow \\ |\mathbb{R}| > \aleph_0 \end{gathered}$$

$\square$

2.4 Metric Spaces

A function $d : X \times X \to \mathbb{R}$ is called a **metric** on X if for all $p, q, r \in X$,

(A) $d(p, q) \geq 0$

(B) $d(p, q) = 0 \iff p = q$

(C) $d(p, q) = d(q, p)$

(D) $d(p, r) \leq d(p, q) + d(q, r)$

(D) is called the **triangle inequality**. The pair (X, d) is called a **metric space**.

Theorem 2.4.1. *Let $n \in \mathbb{N}$ be given. Define $d : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ as*

$$d(\boldsymbol{x}, \boldsymbol{y}) := \|\boldsymbol{x} - \boldsymbol{y}\| = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}$$

*for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. Then d is a metric on $\mathbb{R}^n$. In particular, d is called the **standard metric** on $\mathbb{R}^n$.*

Proof. Let $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^n$ be given.

- $d(\boldsymbol{x}, \boldsymbol{y}) \geq 0$ is trivial.
- By Definition 1.9.1,

$$\begin{aligned} d(\boldsymbol{x}, \boldsymbol{y}) = 0 &\iff \|\boldsymbol{x} - \boldsymbol{y}\| = 0 \\ &\iff \forall k \in \mathbb{N}^n, x_k = y_k \\ &\iff \boldsymbol{x} = \boldsymbol{y} \end{aligned}$$

- $d(\boldsymbol{x}, \boldsymbol{y}) = d(\boldsymbol{y}, \boldsymbol{x})$ is trivial.
- By Theorem 1.9.2,

$$\begin{aligned} d(\boldsymbol{x}, \boldsymbol{z}) &= \|\boldsymbol{x} - \boldsymbol{z}\| = \|(\boldsymbol{x} - \boldsymbol{y}) + (\boldsymbol{y} - \boldsymbol{z})\| \\ &\leq \|\boldsymbol{x} - \boldsymbol{y}\| + \|\boldsymbol{y} - \boldsymbol{z}\| = d(\boldsymbol{x}, \boldsymbol{y}) + d(\boldsymbol{y}, \boldsymbol{z}) \end{aligned}$$

Therefore, d is a metric on $\mathbb{R}^n$. □

Definition 2.4.2. Let $k \in \mathbb{N}$ be given. A set $E \subseteq \mathbb{R}^k$ is called a k**-cell** if

$$E = \left\{ \boldsymbol{x} \in \mathbb{R}^k \;\middle|\; \forall j \in \mathbb{N}^k, a_j \leq x_j \leq b_j \right\}$$

where

$$\forall j \in \mathbb{N}^k, -\infty < a_j \leq b_j < +\infty$$

Definition 2.4.3. Let (X, d) be a metric space. Let $p \in X$ and $r \in \mathbb{R}^+$ be given.

- The **neighborhood** of p with radius r is defined as

$$N_r(p) \coloneqq \{q \in X \mid d(p, q) < r\}$$

- The **deleted neighborhood** of p with radius r is defined as

$$\widetilde{N}_r(p) \coloneqq N_r(p) \setminus \{p\} = \{q \in X \mid 0 < d(p, q) < r\}$$

Definition 2.4.4. Let (X, d) be a metric space and $E \subseteq X$. Let $p \in X$ be given.

- p is called an **limit point** of E if

$$\forall r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) \neq \varnothing$$

  The set of all limit points of E is denoted by E'.

- p is called a **isolated point** of E if

$$\exists r \in \mathbb{R}^+, E \cap N_r(p) = \{p\}$$

- p is called an **interior point** of E if

$$\exists r \in \mathbb{R}^+, N_r(p) \subseteq E$$

Definition 2.4.5. Let (X, d) be a metric space and $E \subseteq X$.

- E is said to be **open** if every point of E is an interior point of E.
- E is said to be **closed** if every limit point of E belongs to E.

Definition 2.4.6. Let (X, d) be a metric space and $E \subseteq X$.

- E is said to be **bounded** if

$$\exists p \in X, \exists r \in \mathbb{R}^+, E \subseteq N_r(p)$$

- E is said to be **perfect** if every point of E is a limit point of E.
- E is said to be **dense** if every point of X is either a limit point of E or a point of E.

Theorem 2.4.7. *Let (X, d) be a metric space.*

$$\forall p \in X, \forall r \in \mathbb{R}^+, N_r(p) \textit{ is open}$$

Proof. Let $q \in N_r(p)$ be given. Then

$$\begin{gathered} d(p,q) < r \\ \Downarrow \\ \varepsilon := r - d(p,q) > 0 \end{gathered}$$

Let $s \in N_\varepsilon(q)$ be given. Then

$$\begin{gathered} d(q,s) < \varepsilon \\ \Downarrow \\ d(p,s) \leq d(p,q) + d(q,s) < d(p,q) + \varepsilon = r \\ \Downarrow \\ s \in N_r(p) \\ \Downarrow \\ N_\varepsilon(q) \subseteq N_r(p) \end{gathered}$$

Since q is arbitrary, $N_r(p)$ is open. □

Theorem 2.4.8. *Let (X, d) be a metric space and $E \subseteq X$.*

$$\forall p \in E', \forall r \in \mathbb{R}^+, |E \cap N_r(p)| \geq \aleph_0$$

Proof. Assume that

$$\exists p \in E', \exists r \in \mathbb{R}^+, |E \cap N_r(p)| < \aleph_0$$

Define

$$S := (E \cap N_r(p)) \setminus \{p\} = E \cap \widetilde{N}_r(p) \subseteq E \cap N_r(p)$$

Then

$$\begin{gathered} p \in E' \\ \Downarrow \\ S \neq \varnothing \end{gathered}$$

By Theorem 2.2.2,

$$\begin{gathered} |E \cap N_r(p)| < \aleph_0 \\ \Downarrow \\ |S| < \aleph_0 \end{gathered}$$

Define

$$T := \{d(p,q) \mid q \in S\} \neq \varnothing$$

Then

$$\begin{gathered} p \notin S \\ \Downarrow \\ 0 \notin T \end{gathered}$$

Define $f : S \twoheadrightarrow T$ as

$$f(q) = d(p, q)$$

By Theorem 2.2.4,

$$|T| < \aleph_0$$

By Theorem 2.2.5,

$$\exists \varepsilon \in \mathbb{R}^+, \varepsilon = \min T$$

Then

$$\begin{gathered} N_\varepsilon(p) \subseteq N_r(p) \\ \Downarrow \\ \forall q \in S, d(p, q) < \varepsilon \\ \Downarrow \\ E \cap \widetilde{N}_\varepsilon(p) = \varnothing \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$|E \cap N_r(p)| \geq \aleph_0$$

□

Theorem 2.4.9. *Let (X, d) be a metric space and $E \subseteq X$. Then*

$$E \textit{ is open} \iff E^c \textit{ is closed}$$

Proof. ($\Longrightarrow$) Let E be open. Let $p \in (E^c)'$ be given. Then

$$\forall r \in \mathbb{R}^+, E^c \cap \widetilde{N}_r(p) \neq \varnothing$$

If $p \in E$, then

$$\begin{gathered} p \in E \\ \Downarrow \\ \exists \varepsilon \in \mathbb{R}^+, N_\varepsilon(p) \subseteq E \\ \Downarrow \\ E^c \cap \widetilde{N}_\varepsilon(p) = \varnothing \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\begin{gathered} p \notin E \\ \Downarrow \\ p \in E^c \end{gathered}$$

Since p is arbitrary,

$$\begin{gathered}(E^c)' \subseteq E^c \\ \Downarrow \\ E^c \text{ is closed}\end{gathered}$$

($\Longleftarrow$) Let E^c be closed. Let $p \in E$ be given. If

$$\nexists r \in \mathbb{R}^+, N_r(p) \subseteq E,$$

then

$$\forall r \in \mathbb{R}^+, E^c \cap N_r(p) \neq \varnothing$$

Since $p \in E$,

$$\begin{gathered}p \notin E^c \\ \Downarrow \\ E^c \cap \widetilde{N}_r(p) \neq \varnothing \\ \Downarrow \\ p \in (E^c)'\end{gathered}$$

Since E^c is closed,

$$\begin{gathered}p \in (E^c)' \\ \Downarrow \\ p \in E^c \\ \Downarrow \\ \perp\end{gathered}$$

Therefore,

$$\begin{gathered}\exists r \in \mathbb{R}^+, N_r(p) \subseteq E \\ \Downarrow \\ p \text{ is an interior point of } E\end{gathered}$$

Since p is arbitrary, E is open. $\square$

Corollary 2.4.10. *Let (X, d) be a metric space and $E \subseteq X$. Then*

$$E \textit{ is closed} \iff E^c \textit{ is open}$$

Theorem 2.4.11. *Let (X, d) be a metric space.*

$$\forall Y, (\forall y \in Y, E_y \subseteq X \textit{ is open}) \implies \bigcup_{y \in Y} E_y \textit{ is open}$$

Proof. Let $p \in \bigcup_{y \in Y} E_y$ be given. Then

$$\exists y_0 \in Y, p \in E_{y_0}$$

Since E_{y0} is open,

$$\begin{gathered}\exists r \in \mathbb{R}^+, N_r(p) \subseteq E_{y0} \\ \Downarrow \\ N_r(p) \subseteq E_{y0} \subseteq \bigcup_{y \in Y} E_y \\ \Downarrow \\ p \text{ is an interior point of } \bigcup_{y \in Y} E_y\end{gathered}$$

Since p is arbitrary, $\bigcup_{y \in Y} E_y$ is open. □

Corollary 2.4.12. *Let (X, d) be a metric space.*

$$\forall Y, (\forall y \in Y, E_y \subseteq X \textit{ is closed}) \implies \bigcap_{y \in Y} E_y \textit{ is closed}$$

Proof. By Corollary 2.4.10,

$$\forall y \in Y, (E_y)^c \text{ is open}$$

By Theorem 2.4.11,

$$\bigcup_{y \in Y} (E_y)^c \text{ is open}$$

By Theorem 2.4.9,

$$\bigcap_{y \in Y} E_y = \left(\bigcup_{y \in Y} (E_y)^c \right)^c \text{ is closed}$$

□

Theorem 2.4.13. *Let (X, d) be a metric space.*

$$(\forall k \in \mathbb{N}^n, E_k \subseteq X \textit{ is open}) \implies \bigcap_{k=1}^{n} E_k \textit{ is open}$$

Proof. Let $p \in \bigcap_{k=1}^{n} E_k$ be given. Then

$$\forall j \in \mathbb{N}^n, p \in E_j$$

Since each E_j is open,

$$\exists r_j \in \mathbb{R}^+, N_{r_j}(p) \subseteq E_j$$

for each $j \in \mathbb{N}^n$. Define

$$R \coloneqq \{r_j \mid j \in \mathbb{N}^n\} \subseteq \mathbb{R}^+$$

Define $f : \mathbb{N}^n \twoheadrightarrow R$ as

$$f(j) = r_j$$

By Lemma 2.2.1 and theorem 2.2.4,

$$|R| < \aleph_0$$

By Theorem 2.2.5,

$$\exists r \in \mathbb{R}^+, r = \min R$$

Then

$$\begin{gathered} \forall j \in \mathbb{N}^n, N_r(p) \subseteq N_{r_j}(p) \subseteq E_j \\ \Downarrow \\ N_r(p) \subseteq \bigcap_{k=1}^{n} E_k \\ \Downarrow \\ p \text{ is an interior point of } \bigcap_{k=1}^{n} E_k \end{gathered}$$

Since p is arbitrary, $\bigcap_{k=1}^{n} E_k$ is open. □

Corollary 2.4.14. *Let (X, d) be a metric space.*

$$(\forall k \in \mathbb{N}^n, E_k \subseteq X \textit{ is closed}) \implies \bigcup_{k=1}^{n} E_k \textit{ is closed}$$

Proof. By Corollary 2.4.10,

$$\forall k \in \mathbb{N}^n, (E_k)^c \text{ is open}$$

By Theorem 2.4.13,

$$\bigcap_{k=1}^{n} (E_k)^c \text{ is open}$$

By Theorem 2.4.9,

$$\bigcup_{k=1}^{n} E_k = \left(\bigcap_{k=1}^{n} (E_k)^c\right)^c \text{ is closed}$$

□

Definition 2.4.15. Let (X, d) be a metric space and $E \subseteq X$. The **closure** $\overline{E}$ of E is defined as

$$\overline{E} := E \cup E'$$

Theorem 2.4.16. *Let (X, d) be a metric space and $E \subseteq X$. Then*

(A) *$\overline{E}$ is closed*

(B) *$E = \overline{E} \iff E$ is closed*

(C) *If $F \subseteq X$ satisfies*

- *$E \subseteq F$*
- *F is closed*

then $\overline{E} \subseteq F$

Proof. **(A)** Let $p \in \left(\overline{E}\right)^c$ be given. Then

$$\begin{gathered} p \notin E \wedge p \notin E' \\ \Downarrow \\ \exists r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) = \varnothing \\ \Downarrow \\ E \cap N_r(p) = \varnothing \\ \Downarrow \\ N_r(p) \subseteq E^c \end{gathered}$$

By Theorem 2.4.7, $N_r(p)$ is open. Then

$$\begin{gathered} \forall q \in N_r(p), \exists \varepsilon \in \mathbb{R}^+, N_\varepsilon(q) \subseteq N_r(p) \subseteq E^c \\ \Downarrow \\ E \cap N_\varepsilon(q) = \varnothing \\ \Downarrow \\ E \cap \widetilde{N}_\varepsilon(q) = \varnothing \\ \Downarrow \\ q \notin E' \end{gathered}$$

Since q is arbitrary,

$$\begin{gathered} N_r(p) \subseteq \left(E'\right)^c \\ \Downarrow \\ N_r(p) \subseteq \left(\overline{E}\right)^c = E^c \cap \left(E'\right)^c \\ \Downarrow \\ p \text{ is an interior point of } \left(\overline{E}\right)^c \end{gathered}$$

Since p is arbitrary,

$$\left(\overline{E}\right)^c \text{ is open}$$

By Corollary 2.4.10,

$$\overline{E} \text{ is closed}$$

(B) E is closed $\iff E' \subseteq E \iff E \cup E' = E \iff \overline{E} = E$

(C) Let $F \subseteq X$ satisfy

- $E \subseteq F$
- F is closed

Let $p \in E'$ be given. Then,

$$\begin{gathered} \forall r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) \neq \varnothing \\ \Downarrow \\ F \cap \widetilde{N}_r(p) \neq \varnothing \\ \Downarrow \\ p \in F' \subseteq F \end{gathered}$$

Since p is arbitrary,

$$\begin{gathered} E' \subseteq F' \subseteq F \\ \Downarrow \\ \overline{E} = E \cup E' \subseteq F \end{gathered}$$

$\square$

Theorem 2.4.17. *Let $(\mathbb{R}, \|\cdot\|)$ be given. If $E \subseteq \mathbb{R}$ satisfies*

- $E \neq \varnothing$
- *E is bounded above*

then

$$\exists \sup E \in \overline{E}$$

Also,

$$E \textit{ is closed} \implies \exists \sup E \in E$$

Proof. By the *least upper bound property*,

$$\exists \alpha \in \mathbb{R}, \alpha = \sup E$$

If $\alpha \in E$, then

$$\begin{gathered} E \subseteq \overline{E} \\ \Downarrow \\ \alpha \in \overline{E} \end{gathered}$$

Otherwise,

$$\begin{gathered} \forall x \in \mathbb{R}^+, \alpha - x \text{ is not an upper bound of } E \\ \Downarrow \\ \exists p \in E, \alpha - x < p < \alpha \\ \Downarrow \\ p \in \widetilde{N}_x(\alpha) \cap E \\ \Downarrow \\ \alpha \in E' \end{gathered}$$

Therefore,

$$\begin{gathered} E' \subseteq \overline{E} \\ \Downarrow \\ \alpha \in \overline{E} \end{gathered}$$

□

Theorem 2.4.18. *Let (X,d) be a metric space and $E \subseteq X$. Then*

$$E \setminus E' = \{p \in X \mid p \textit{ is an isolated point of } E\}$$

Proof. **($\subseteq$)** Let $p \in E \setminus E'$ be given. Then

$$\begin{gathered} p \notin E' \\ \Downarrow \\ \exists r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) = \varnothing \\ \Downarrow \\ (E \cap N_r(p)) \setminus \{p\} = \varnothing \end{gathered}$$

Therefore,

$$\begin{gathered} p \in E \wedge p \in N_r(p) \\ \Downarrow \\ E \cap N_r(p) = \{p\} \\ \Downarrow \\ p \text{ is an isolated point of } E \end{gathered}$$

($\supseteq$) Let p be an isolated point of E. Then

$$\exists r \in \mathbb{R}^+, E \cap N_r(p) = \{p\}$$

Therefore,

$$\begin{gathered} p \in E \\ \Downarrow \\ E \cap \widetilde{N}_r(p) = \varnothing \\ \Downarrow \\ p \notin E' \\ \Downarrow \\ p \in E \setminus E' \end{gathered}$$

□

Definition 2.4.19. Let (X,d) be a metric space and $E \subseteq X$. The **boundary** ∂E of E is defined as

$$\partial E := \overline{E} \cap \overline{E^c}$$

Lemma 2.4.20. *Let* $(\mathbb{R}^n, \|\cdot\|)$ *be given for* $n \in \mathbb{N}$. *Then*

$$\forall \boldsymbol{p} \in \mathbb{R}^n, \forall r \in \mathbb{R}^+, \widetilde{N}_r(\boldsymbol{p}) \neq \varnothing$$

Proof. Let $\boldsymbol{p} \in \mathbb{R}^n$ and $r \in \mathbb{R}^+$ be given. Then

$$d\Big(\boldsymbol{p} + \frac{r}{2}\boldsymbol{e_1}, \boldsymbol{p}\Big) = \Big\|\Big(\boldsymbol{p} + \frac{r}{2}\boldsymbol{e_1}\Big) - \boldsymbol{p}\Big\| = \Big\|\frac{r}{2}\boldsymbol{e_1}\Big\| = \frac{r}{2}\|\boldsymbol{e_1}\| = \frac{r}{2} < r$$

where $\boldsymbol{e_1} = (1, 0, 0, \dots, 0) \in \mathbb{R}^n$. Therefore,

$$\begin{gathered} \boldsymbol{p} \neq \Big(\boldsymbol{p} + \frac{r}{2}\boldsymbol{e_1}\Big) \in \widetilde{N}_r(\boldsymbol{p}) \\ \Downarrow \\ \widetilde{N}_r(\boldsymbol{p}) \neq \varnothing \end{gathered}$$

□

Theorem 2.4.21. *Let* $(\mathbb{R}^n, \|\cdot\|)$ *be given for* $n \in \mathbb{N}$ *and* $E \subseteq \mathbb{R}^n$. *Then*

$$\partial E = \big(E \setminus E'\big) \cup \big(E^c \setminus (E^c)'\big) \cup \big(E' \cap (E^c)'\big)$$

This states that ∂E *consists of*

- *The set of all isolated points of* E
- *The set of all isolated points of* E^c
- *The set of all limit points of both* E *and* E^c

where the three sets are disjoint.

Proof. By definition,

$$\begin{aligned} \partial E &= \overline{E} \cap \overline{E^c} \\ &= \big(E \cup E'\big) \cap \big(E^c \cup (E^c)'\big) \\ &= (E \cap E^c) \cup \big(E \cap (E^c)'\big) \cup \big(E' \cap E^c\big) \cup \big(E' \cap (E^c)'\big) \end{aligned}$$

Consider the following equivalences:

- $E = \big(E \setminus E'\big) \cup \big(E \cap E'\big)$
- $E^c = \big(E^c \setminus (E^c)'\big) \cup \big(E^c \cap (E^c)'\big)$

Then

- $E \cap (E^c)' = \big(\big(E \setminus E'\big) \cap (E^c)'\big) \cup \big(\big(E \cap E'\big) \cap (E^c)'\big)$
- $E' \cap E^c = \big(E' \cap \big(E^c \setminus (E^c)'\big)\big) \cup \big(E' \cap \big(E^c \cap (E^c)'\big)\big)$

Consider the last two terms of the above equalities:

- $\big((E \cap E'\big) \cap (E^c)'\big) \subseteq E' \cap (E^c)'$
- $\big(E' \cap \big(E^c \cap (E^c)'\big)\big) \subseteq E' \cap (E^c)'$

Then

$$\begin{aligned}\partial E &= \varnothing \cup \big(\big(E \setminus E'\big) \cap (E^c)'\big) \cup \big(E' \cap \big(E^c \setminus (E^c)'\big)\big) \cup \big(E' \cap (E^c)'\big) \\ &= \big(\big(E \setminus E'\big) \cap (E^c)'\big) \cup \big(E' \cap \big(E^c \setminus (E^c)'\big)\big) \cup \big(E' \cap (E^c)'\big)\end{aligned}$$

Also,

$$\begin{gathered}\forall \boldsymbol{p} \in E \setminus E', \exists r \in \mathbb{R}^+, E \cap \widetilde{N}_r(\boldsymbol{p}) = \varnothing \\ \Downarrow \\ \widetilde{N}_r(\boldsymbol{p}) \subseteq E^c\end{gathered}$$

By Lemma 2.4.20,

$$\forall s \in \mathbb{R}^+, \widetilde{N}_s(\boldsymbol{p}) \cap \widetilde{N}_r(\boldsymbol{p}) = \widetilde{N}_{\min(s,r)}(\boldsymbol{p}) \neq \varnothing$$

Then

$$\begin{gathered}\forall s \in \mathbb{R}^+, \widetilde{N}_s(\boldsymbol{p}) \cap \widetilde{N}_r(\boldsymbol{p}) \neq \varnothing \\ \Downarrow \\ \widetilde{N}_s(\boldsymbol{p}) \cap \left(\widetilde{N}_r(\boldsymbol{p}) \cap E^c\right) \neq \varnothing \\ \Downarrow \\ \varnothing \neq \left(\widetilde{N}_s(\boldsymbol{p}) \cap E^c\right) \cap \widetilde{N}_r(\boldsymbol{p}) \subseteq \widetilde{N}_s(\boldsymbol{p}) \cap E^c \\ \Downarrow \\ E^c \cap \widetilde{N}_s(\boldsymbol{p}) \neq \varnothing \\ \Downarrow \\ \boldsymbol{p} \in (E^c)'\end{gathered}$$

Then

$$\begin{gathered}E \setminus E' \subseteq (E^c)' \\ \Downarrow \\ \big(E \setminus E'\big) \cap (E^c)' = E \setminus E'\end{gathered}$$

Similarly,

$$\begin{gathered}E^c \setminus (E^c)' \subseteq E' \\ \Downarrow \\ E' \cap \big(E^c \setminus (E^c)'\big) = E^c \setminus (E^c)'\end{gathered}$$

Therefore,

$$\partial E = \big(E \setminus E'\big) \cup \big(E^c \setminus (E^c)'\big) \cup \big(E' \cap (E^c)'\big)$$

$\square$

Remark. Let (X, d) be a metric space and $Y \subseteq X$. Then

$$d|_{Y\times Y} : Y \times Y \to \mathbb{R}$$

satisfies the properties of a metric. Therefore, the pair

$$(Y, d|_{Y\times Y})$$

is also a metric space, which is called the **subspace** of (X, d). Since

- $d|_{Y\times Y}(p, q)$
- $(Y, d|_{Y\times Y})$

are cumbersome, we will write

- $d(p, q)$ instead of $d|_{Y\times Y}(p, q)$
- (Y, d) instead of $(Y, d|_{Y\times Y})$

for the sake of simplicity.

Definition 2.4.22. Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. E is called **open relative** to Y if E is open in (Y, d). That is,

$$\forall p \in E, \exists r \in \mathbb{R}^+, Y \cap N_r(p) \subseteq E$$

Theorem 2.4.23. *Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. If $V \subseteq X$ satisfies*

- *V is open in (X, d)*
- *$E = V \cap Y$*

then E is open relative to Y. The converse is also true.

Proof. **($\Longrightarrow$)** Let $V \subseteq X$ satisfies

- V is open in (X, d)
- $E = V \cap Y$

Then

$$\begin{gathered}\forall p \in E, \exists r_p \in \mathbb{R}^+, N_{r_p}(p) \subseteq V \\ \Downarrow \\ Y \cap N_{r_p}(p) \subseteq V \cap Y = E\end{gathered}$$

($\Longleftarrow$) Let E be open relative to Y. Then

$$\forall p \in E, \exists r_p \in \mathbb{R}^+, Y \cap N_{r_p}(p) \subseteq E$$

Define

$$V := \bigcup_{p \in E} N_{r_p}(p)$$

By Theorems 2.4.7 and 2.4.11, V is open in (X, d).

$$\begin{gathered} \forall p \in E, p \in N_{r_p}(p) \subseteq V \\ \Downarrow \\ E \subseteq Y \wedge E \subseteq V \\ \Downarrow \\ E \subseteq V \cap Y \end{gathered}$$

Also,

$$\begin{gathered} \forall p \in E, Y \cap N_{r_p}(p) \subseteq E \\ \Downarrow \\ \bigcup_{p \in E} \big(Y \cap N_{r_p}(p)\big) \subseteq E \\ \Downarrow \\ Y \cap \bigcup_{p \in E} N_{r_p}(p) \subseteq E \\ \Downarrow \\ Y \cap V \subseteq E \end{gathered}$$

Therefore,

$$E = V \cap Y$$

□

2.5 Compact Sets

Let (X, d) be a metric space and $E \subseteq X$. Let A be a set and let G_α be an open set in X for each $\alpha \in A$. Then the collection $\{G_\alpha\}_{\alpha \in A}$ is called an **open cover** of E if

$$E \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

A collection $\{G_\alpha\}_{\alpha \in S}$ is called a **subcover** of E if

- $S \subseteq A$.
- $E \subseteq \bigcup\limits_{\alpha \in S} G_\alpha$.

E is called **compact** if every open cover of E has a finite subcover. That is, for each open cover $\{G_\alpha\}_{\alpha \in A}$ of E, there exists a subset $S_A \subseteq A$ such that

- S_A is finite.

- $E \subseteq \bigcup_{\alpha \in S_A} G_\alpha$.

Theorem 2.5.1. *Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. Then E is compact relative to Y if and only if E is compact relative to X.*

Proof. ($\Longrightarrow$) Let E be a compact set relative to Y. Let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of E relative to X. Then we have

$$E \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

Let $H_\alpha = G_\alpha \cap Y$ for each $\alpha \in A$. By Theorem 2.4.23, H_α is open relative to Y. Since $E \subseteq Y$, we have

$$E \subseteq \left(\bigcup_{\alpha \in A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in A} (G_\alpha \cap Y) = \bigcup_{\alpha \in A} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in A}$ is an open cover of E relative to Y. Since E is compact relative to Y, there exists a finite subcover $\{H_\alpha\}_{\alpha \in S_A}$ of E relative to Y. Then we have

$$E \subseteq \bigcup_{\alpha \in S_A} H_\alpha = \bigcup_{\alpha \in S_A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in S_A} G_\alpha.$$

Thus $\{G_\alpha\}_{\alpha \in S_A}$ is a finite subcover of E relative to X. Therefore, E is compact relative to X.

($\Longleftarrow$) Let E be a compact set relative to X. Let $\{H_\alpha\}_{\alpha \in A}$ be an open cover of E relative to Y. Then we have

$$E \subseteq \bigcup_{\alpha \in A} H_\alpha.$$

By Theorem 2.4.23, there exists an open set G_α in X such that $H_\alpha = G_\alpha \cap Y$ for each $\alpha \in A$. Then we have

$$E \subseteq \bigcup_{\alpha \in A} H_\alpha = \bigcup_{\alpha \in A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

Thus $\{G_\alpha\}_{\alpha \in A}$ is an open cover of E relative to X. Since E is compact relative to X, there exists a finite subcover $\{G_\alpha\}_{\alpha \in S_A}$ of E relative to X. Then we have

$$E \subseteq \left(\bigcup_{\alpha \in S_A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in S_A} (G_\alpha \cap Y) = \bigcup_{\alpha \in S_A} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in S_A}$ is a finite subcover of E relative to Y. Therefore, E is compact relative to Y. $\square$

Remark. The concept of **compactness** is preserved under the subspace topology, while the concepts of **openness** and **closedness** are not, which is the reason why

"Let (X, d) be a metric space and $E \subseteq X$."

this kind of statement is always included in theorems and definitions about metric spaces. In this textbook, you can assume that

- E is open. $\implies$ E is open relative to X.
- E is closed. $\implies$ E is closed relative to X.
- E is compact. $\implies$ E is compact relative to X.

if there is no specific mention of the subspace topology.

Theorem 2.5.2. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then E is closed.*

Proof. Let E be compact. We will prove that E^c is open, which implies that E is closed by Corollary 2.4.10. Let $p \in E^c$ be given. Since $p \notin E$, we have

$$r_q := \frac{d(p,q)}{2} > 0 \quad \text{for each } q \in E.$$

Since $q \in N_{r_q}(q)$ for each $q \in E$, we have

$$E \subseteq \bigcup_{q \in E} N_{r_q}(q) \implies \left\{N_{r_q}(q)\right\}_{q \in E} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\left\{N_{r_q}(q)\right\}_{q \in S}$ of E. Let T be a set defined as

$$T := \{r_q \mid q \in S\}.$$

Let $f : S \to T$ be a function defined as

$$f(q) = r_q.$$

Then f is surjective. By Theorem 2.2.4, T is finite. Then, by Theorem 2.2.5, T has a minimum element, say $\varepsilon = \min T > 0$. For every $x \in N_\varepsilon(p)$, we have

$$d(x,p) < \varepsilon \leq r_q \quad \text{for each } q \in S.$$

Assume that there exists some $q \in S$ such that $x \in N_{r_q}(q)$. Then we have

$$d(p,q) \leq d(p,x) + d(x,q) < r_q + r_q = 2r_q = d(p,q),$$

which is a contradiction. Then the following statements hold:

- $N_\varepsilon(p) \cap N_{r_q}(q) = \varnothing$ for each $q \in S$.
- $E \subseteq \bigcup_{q \in S} N_{r_q}(q)$.

Therefore, we have $N_\varepsilon(p) \subseteq E^c$. Since p is arbitrary, E^c is open. $\square$

Theorem 2.5.3. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then E is bounded.*

Proof. Let E be compact. Let $p \in E$ be given. Let $\{G_n\}_{n\in\mathbb{N}}$ be a collection of open sets defined as

$$G_n = N_n(p) \quad \text{for each } n \in \mathbb{N}.$$

Then we have

$$n \leq m \implies G_n \subseteq G_m \quad \text{for each } n, m \in \mathbb{N}.$$

For every $q \in X$, by Theorem 1.6.1, there exists $n \in \mathbb{N}$ such that $n > d(p, q)$. Thus we have

$$E \subseteq X \subseteq \bigcup_{n=1}^{\infty} G_n \implies \{G_n\}_{n\in\mathbb{N}} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{G_n\}_{n\in S}$ of E. Since $S \subseteq \mathbb{N}$ is finite, by Theorem 2.2.5, S has a greatest element, say $m = \max S$. Then we have

$$E \subseteq \bigcup_{n\in S} G_n \subseteq G_m = N_m(p),$$

which implies that E is bounded. $\square$

Theorem 2.5.4. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then every closed subset $F \subseteq E$ is compact.*

Proof. Let $\{H_\alpha\}_{\alpha\in A}$ be an open cover of F. We have the following statements:

- F is closed. $\implies$ F^c is open.
- $F \subseteq \bigcup_{\alpha\in A} H_\alpha$.

Then we have

- $X = F \cup F^c \subseteq \left(\bigcup_{\alpha\in A} H_\alpha\right) \cup F^c = \bigcup_{\alpha\in A} (H_\alpha \cup F^c)$.

- $E \subseteq X \implies E \subseteq \bigcup_{\alpha \in A} (H_\alpha \cup F^c)$.

By Theorem 2.4.11, $H_\alpha \cup F^c$ is open for each $\alpha \in A$. Thus

$$\{H_\alpha \cup F^c\}_{\alpha \in A} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{H_\alpha \cup F^c\}_{\alpha \in B}$ of E. Then we have

- $E \subseteq \bigcup_{\alpha \in B} (H_\alpha \cup F^c)$.
- $F \subseteq E \implies F \subseteq \bigcup_{\alpha \in B} (H_\alpha \cup F^c) = \left(\bigcup_{\alpha \in B} H_\alpha \right) \cup F^c$.

Since $F \cap F^c = \varnothing$, we have

$$F \subseteq \bigcup_{\alpha \in B} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in B}$ is a finite subcover of F. Therefore, F is compact. □

Theorem 2.5.5. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then every infinite subset of E has a limit point in E.*

Proof. Let E be compact. Let $F \subseteq E$ be an infinite subset of E. Assume that F has no limit point in E. Then, for each $p \in E$, there exists $r_p \in \mathbb{R}^+$ such that

$$\widetilde{N}_{r_p}(p) \cap F = \varnothing \implies N_{r_p}(p) \cap F \subseteq \{p\}.$$

Let $\{G_\alpha\}_{\alpha \in E}$ be a collection of open sets defined as

$$G_\alpha = N_{r_\alpha}(\alpha) \quad \text{for each } \alpha \in E.$$

Since $\alpha \in G_\alpha$ for each $\alpha \in E$, we have

$$E \subseteq \bigcup_{\alpha \in E} G_\alpha \implies \{G_\alpha\}_{\alpha \in E} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{G_\alpha\}_{\alpha \in S}$ of E. Then we have

$$E \subseteq \bigcup_{\alpha \in S} G_\alpha \implies F \subseteq E \subseteq \bigcup_{\alpha \in S} G_\alpha.$$

Since $F \subseteq \bigcup_{\alpha \in S} G_\alpha$, we have

$$F = F \cap \bigcup_{\alpha \in S} G_\alpha = \bigcup_{\alpha \in S} (F \cap G_\alpha) \subseteq \bigcup_{\alpha \in S} \{\alpha\} = S,$$

which is a contradiction by Theorem 2.2.2. Therefore, every infinite subset of E has a limit point in E. □

Theorem 2.5.6. *Let (X, d) be a metric space and let $\{K_\alpha\}_{\alpha \in A}$ be a collection of compact sets. Then we have*

$$\bigcap_{\alpha \in B} K_\alpha \neq \varnothing \quad \textit{for all finite subset } B \subseteq A \implies \bigcap_{\alpha \in A} K_\alpha \neq \varnothing.$$

Proof. Assume that $\bigcap_{\alpha \in A} K_\alpha = \varnothing$. Then we have

$$X = \left(\bigcap_{\alpha \in A} K_\alpha\right)^c = \bigcup_{\alpha \in A} (K_\alpha)^c.$$

Since each K_α is compact, K_α is closed by Theorem 2.5.2. Then, by Corollary 2.4.10, we have

$$\text{Each } (K_\alpha)^c \text{ is open} \implies \{(K_\alpha)^c\}_{\alpha \in A} \text{ is an open cover of } X.$$

Since $K_\alpha \subseteq X \subseteq \bigcup_{\beta \in A} (K_\beta)^c$, we have

$$\{(K_\beta)^c\}_{\beta \in A} \text{ is an open cover of } K_\alpha.$$

Since K_α is compact, there exists a finite subcover $\{(K_\beta)^c\}_{\beta \in B_\alpha}$ of K_α. Then we have

$$\begin{aligned} K_\alpha \subseteq \bigcup_{\beta \in B_\alpha} (K_\beta)^c = \left(\bigcap_{\beta \in B_\alpha} K_\beta\right)^c \\ \implies K_\alpha \cap \left(\bigcap_{\beta \in B_\alpha} K_\beta\right) = \varnothing \\ \implies \bigcap_{\beta \in B_\alpha \cup \{\alpha\}} K_\beta = \varnothing. \end{aligned}$$

Since $\{\alpha\}$ and B_α are finite, $B_\alpha \cup \{\alpha\}$ is also finite by Theorem 2.2.11, which contradicts the assumption. Therefore, we have

$$\bigcap_{\alpha \in A} K_\alpha \neq \varnothing.$$

□

Lemma 2.5.7. *Let $(a_n)_{n=1}^\infty$ and $(b_n)_{n=1}^\infty$ be sequences in $\mathbb{R}$, and let $\{I_n\}_{n=1}^\infty$ be a sequence of intervals in $\mathbb{R}$ such that*

- $I_n = [a_n, b_n]$ *for all* $n \in \mathbb{N}$
- $I_{n+1} \subseteq I_n \iff a_n \leq a_{n+1} \leq b_{n+1} \leq b_n$ *for all* $n \in \mathbb{N}$

where $a_n \leq b_n$ *for all* $n \in \mathbb{N}$*. Then we have*

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing.$$

Proof. Let E be a set defined as

$$E \coloneqq \{a_n \mid n \in \mathbb{N}\}.$$

Since E is bounded above by b_1, there exists $\alpha = \sup E$. Then we have

$$a_m \leq \alpha \quad \text{for all } m \in \mathbb{N}.$$

For every $m \in \mathbb{N}$, we have b_m is an upper bound of E since

- $a_n \leq a_m \leq b_m$ for all $n \in \mathbb{N}$ such that $n \leq m$.
- $a_n \leq b_n \leq b_m$ for all $n \in \mathbb{N}$ such that $n > m$.

Therefore, we have

$$a_m \leq \alpha \leq b_m \implies \alpha \in I_m \text{ for all } m \in \mathbb{N}.$$

□

Theorem 2.5.8. *Let* $k \in \mathbb{N}$ *be given. Let* $\{a_{n,j}\}_{n=1}^{\infty}$ *and* $\{b_{n,j}\}_{n=1}^{\infty}$ *be sequences in* $\mathbb{R}$ *for all* $j = 1, 2, \ldots, k$*, and let* $\{I_n\}_{n=1}^{\infty}$ *be a sequence of* k*-cells in* $\mathbb{R}^k$ *such that*

- $I_n = \left\{\boldsymbol{x} \in \mathbb{R}^k \;\middle|\; a_{n,j} \leq x_j \leq b_{n,j} \text{ for all } j = 1, 2, \ldots, k\right\}$ *for all* $n \in \mathbb{N}$
- $I_{n+1} \subseteq I_n \iff a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j}$ *for all* $j = 1, 2, \ldots, k$ *and* $n \in \mathbb{N}$

where $a_{n,j} \leq b_{n,j}$ *for all* $j = 1, 2, \ldots, k$ *and* $n \in \mathbb{N}$*. Then we have*

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing.$$

Proof. Let $\{I_{n,j}\}_{n=1}^{\infty}$ be a sequence of intervals in $\mathbb{R}$ defined as

$$I_{n,j} = [a_{n,j}, b_{n,j}] \quad \text{for all } j = 1, 2, \ldots, k \text{ and } n \in \mathbb{N}.$$

Then we have $I_{n+1,j} \subseteq I_{n,j}$ for all $j = 1, 2, \ldots, k$ and $n \in \mathbb{N}$. By Lemma 2.5.7, we have

$$\bigcap_{n=1}^{\infty} I_{n,j} \neq \varnothing \quad \text{for all } j = 1, 2, \ldots, k.$$

Let $\alpha_j \in \bigcap_{n=1}^{\infty} I_{n,j}$ be given for all $j = 1, 2, \ldots, k$. Then we have

$$\alpha_j \in I_{n,j} \implies a_{n,j} \leq \alpha_j \leq b_{n,j} \quad \text{for all } j = 1, 2, \ldots, k \text{ and } n \in \mathbb{N}.$$

Let $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \ldots, \alpha_k) \in \mathbb{R}^k$. Then we have

$$\boldsymbol{\alpha} \in I_n \quad \text{for all } n \in \mathbb{N} \implies \boldsymbol{\alpha} \in \bigcap_{n=1}^{\infty} I_n.$$

□

Theorem 2.5.9. *Let* $\left(\mathbb{R}^k, \|\cdot\|\right)$ *be given for* $k \in \mathbb{N}$. *Then every* k*-cell is compact.*

Proof. Let I be a k-cell in $\mathbb{R}^k$, and let $I_1 := I$. Then we have

$$I_1 = \prod_{j=1}^{k} [a_{1,j}, b_{1,j}] \subseteq \mathbb{R}^k,$$

where $-\infty < a_{1,j} \leq b_{1,j} < +\infty$ for all $j = 1, 2, \ldots, k$. Let $I_{1,j,m}$ be an interval in $\mathbb{R}$ defined as

$$I_{1,j,m} := \begin{cases} \left[a_{1,j}, \dfrac{a_{1,j} + b_{1,j}}{2}\right], & \text{if } m = 1 \\ \left[\dfrac{a_{1,j} + b_{1,j}}{2}, b_{1,j}\right], & \text{if } m = 2 \end{cases} \quad \text{for all } j = 1, 2, \ldots, k.$$

Let S_k be a set defined as

$$S_k := \{1, 2\}^k = \left\{\boldsymbol{m} \in \mathbb{R}^k \;\middle|\; m_j \in \{1, 2\} \text{ for all } j = 1, 2, \ldots, k\right\}.$$

By Theorem 2.2.13, S_k is finite. We have

$$I_1 = \bigcup_{\boldsymbol{m} \in S_k} \left(\prod_{j=1}^{k} I_{1,j,m_j}\right).$$

Let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of I_1. Assume that I_1 has no finite subcover. Then, for each $\boldsymbol{m} \in S_k$, we have

$$\prod_{j=1}^{k} I_{1,j,m_j} \subseteq I_1 \subseteq \bigcup_{\alpha \in A} G_\alpha,$$

which implies that $\{G_\alpha\}_{\alpha \in A}$ is an open cover of $\prod_{j=1}^{k} I_{1,j,m_j}$. Assume that for each $\boldsymbol{m} \in S_k$, we have

$$\prod_{j=1}^{k} I_{1,j,m_j} \text{ has a finite subcover } \{G_\alpha\}_{\alpha \in A_{\boldsymbol{m}}}.$$

Then we have

$$\prod_{j=1}^{k} I_{1,j,m_j} \subseteq \bigcup_{\alpha \in A_{\boldsymbol{m}}} G_\alpha \quad \text{for all } \boldsymbol{m} \in S_k,$$

where $\bigcup_{\alpha \in A_{\boldsymbol{m}}} G_\alpha$ is open by Theorem 2.4.11. Then we have

$$I_1 \subseteq \bigcup_{\boldsymbol{m} \in S_k} \left(\bigcup_{\alpha \in A_{\boldsymbol{m}}} G_\alpha \right) = \bigcup_{\alpha \in A} G_\alpha,$$

where $A := \bigcup_{\boldsymbol{m} \in S_k} A_{\boldsymbol{m}}$ is finite by Theorem 2.2.11. Thus I_1 has a finite subcover, which is a contradiction. Therefore, there exists $\boldsymbol{m_1} \in S_k$ such that

$$\prod_{j=1}^{k} I_{1,j,(m_1)_j} \text{ has no finite subcover.}$$

Let I_2 be a k-cell in $\mathbb{R}^k$ defined as

$$I_2 := \prod_{j=1}^{k} [a_{2,j}, b_{2,j}] \subseteq \mathbb{R}^k,$$

where $a_{2,j}, b_{2,j} \in \mathbb{R}$ is defined as

- $a_{2,j} := \begin{cases} a_{1,j}, & \text{if } (m_1)_j = 1 \\ \dfrac{a_{1,j} + b_{1,j}}{2}, & \text{if } (m_1)_j = 2 \end{cases}$
- $b_{2,j} := \begin{cases} \dfrac{a_{1,j} + b_{1,j}}{2}, & \text{if } (m_1)_j = 1 \\ b_{1,j}, & \text{if } (m_1)_j = 2 \end{cases}$

for all $j = 1, 2, \ldots, k$. Then, for each $j = 1, 2, \ldots, k$, we have

$$[a_{2,j}, b_{2,j}] = I_{1,j,(m_1)_j} \subseteq [a_{1,j}, b_{1,j}],$$

which implies that

$$I_2 = \prod_{j=1}^{k} I_{1,j,(m_1)_j} \subseteq I_1.$$

Let $I_{2,j,m}$ be an interval in $\mathbb{R}$ defined as

$$I_{2,j,m} := \begin{cases} \left[a_{2,j}, \dfrac{a_{2,j}+b_{2,j}}{2}\right], & \text{if } m = 1 \\ \left[\dfrac{a_{2,j}+b_{2,j}}{2}, b_{2,j}\right], & \text{if } m = 2 \end{cases} \quad \text{for all } j = 1, 2, \ldots, k.$$

Then we have

$$I_2 = \bigcup_{\boldsymbol{m} \in S_k} \left(\prod_{j=1}^{k} I_{2,j,m_j} \right).$$

We have shown that $\{G_\alpha\}_{\alpha \in A}$ is an open cover of I_2, and I_2 has no finite subcover. Then, for each $\boldsymbol{m} \in S_k$, we have

$$\prod_{j=1}^{k} I_{2,j,m_j} \subseteq I_2 \subseteq \bigcup_{\alpha \in A} G_\alpha,$$

which implies that $\{G_\alpha\}_{\alpha \in A}$ is an open cover of $\displaystyle\prod_{j=1}^{k} I_{2,j,m_j}$. Therefore, there exists $\boldsymbol{m_2} \in S_k$ such that

$$\prod_{j=1}^{k} I_{2,j,(m_2)_j} \text{ has no finite subcover.}$$

By repeating this process, we can construct the following sequences:

- The sequence $(\boldsymbol{m}_n)_{n=1}^{\infty}$ in S_k.
- The sequences $(a_{n,j})_{n=1}^{\infty}$ and $(b_{n,j})_{n=1}^{\infty}$ in $\mathbb{R}$ for all $j = 1, 2, \ldots, k$.
- The sequence $(I_{n,j,m})_{n=1}^{\infty}$ of intervals in $\mathbb{R}$ for all $j = 1, 2, \ldots, k$ and $m = 1, 2$.
- The sequence $(I_n)_{n=1}^{\infty}$ of k-cells in $\mathbb{R}^k$.

We have the following properties:

- $I_{n+1} \subseteq I_n$ for all $n \in \mathbb{N}$.
- I_n has no finite subcover for all $n \in \mathbb{N}$.
- $a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j}$ for all $j = 1, 2, \ldots, k$ and $n \in \mathbb{N}$.
- $b_{n+1,j} - a_{n+1,j} = \dfrac{1}{2}(b_{n,j} - a_{n,j}) \implies b_{n,j} - a_{n,j} = \dfrac{1}{2^{n-1}}(b_{1,j} - a_{1,j})$ for all $j = 1, 2, \ldots, k$ and $n \in \mathbb{N}$.

Let $\delta := \sqrt{\sum_{j=1}^{k}(b_{1,j} - a_{1,j})^2}$. Then we have

$$\|\boldsymbol{x} - \boldsymbol{y}\| \leq \sqrt{\sum_{j=1}^{k}(b_{n,j} - a_{n,j})^2} = \frac{\delta}{2^{n-1}} \quad \text{for all } \boldsymbol{x}, \boldsymbol{y} \in I_n \text{ and } n \in \mathbb{N}.$$

By Theorem 2.5.8, we have $\bigcap_{n=1}^{\infty} I_n \neq \varnothing$. Let $\boldsymbol{\beta} \in \bigcap_{n=1}^{\infty} I_n$ be given. Then there exists $\alpha_0 \in A$ such that $\boldsymbol{\beta} \in G_{\alpha_0}$. Since G_{α_0} is open, there exists $\varepsilon \in \mathbb{R}^+$ such that

$$B_\varepsilon(\boldsymbol{\beta}) \subseteq G_{\alpha_0}.$$

By Corollary 1.6.2, there exists $N \in \mathbb{N}$ such that $\frac{1}{2^N} < \frac{\varepsilon}{2\delta}$. Since $\boldsymbol{\beta} \in I_N$, we have

$$\|\boldsymbol{\beta} - \boldsymbol{\gamma}\| \leq \frac{\delta}{2^{N-1}} < \varepsilon \quad \text{for all } \boldsymbol{\gamma} \in I_N,$$

which implies that $I_N \subseteq B_\varepsilon(\boldsymbol{\beta}) \subseteq G_{\alpha_0}$. Therefore, I_N has a finite subcover $\{G_{\alpha_0}\}$, which is a contradiction. Thus I has a finite subcover, and I is compact. $\square$

Corollary 2.5.10. *For every $-\infty < a \leq b < +\infty$, the closed interval $[a, b]$ is compact in $(\mathbb{R}, \|\cdot\|)$.*

Theorem 2.5.11. *Let $(\mathbb{R}^n, \|\cdot\|)$ be given for $n \in \mathbb{N}$ and $E \subseteq \mathbb{R}^n$. Then we have*

$$E \textit{ is compact} \iff E \textit{ is closed and bounded.}$$

This theorem is called the ***Heine-Borel Theorem****.*

Proof. ($\Longrightarrow$) Let E be compact. E is closed by Theorem 2.5.2, and E is bounded by Theorem 2.5.3.

($\Longleftarrow$) Let E be closed and bounded. Then there exists some $\boldsymbol{p} \in \mathbb{R}^n$ and $r \in \mathbb{R}^+$ such that $E \subseteq B_r(\boldsymbol{p})$. Let I be a n-cell in $\mathbb{R}^n$ defined as

$$I := \prod_{j=1}^{n}[p_j - r, p_j + r] \subseteq \mathbb{R}^n.$$

Then we have $E \subseteq B_r(\boldsymbol{p}) \subseteq I$. By Theorem 2.5.9, I is compact. Since E is closed, by Theorem 2.5.4, E is compact. $\square$

2.6 Connected Sets

Let (X, d) be a metric space and $E \subseteq X$. E is called **connected** if there exists no pair of sets $A, B \subseteq X$ such that

- $\varnothing \neq A \subseteq E$ and $\varnothing \neq B \subseteq E$.
- $\overline{A} \cap B = A \cap \overline{B} = \varnothing$.
- $E = A \cup B$.

Such pair of sets A and B is called **separated**. If E is not connected, then E is called **disconnected**.

Theorem 2.6.1. *Let* $(\mathbb{R}, \|\cdot\|)$ *be given. Then* $E \subseteq \mathbb{R}$ *is connected if and only if*

$$a < x < b \implies x \in E \quad \textit{for all } a, b \in E \textit{ and } x \in \mathbb{R}.$$

Proof. ($\Longrightarrow$) Let $E \subseteq \mathbb{R}$ be connected. Assume that there exist $a, b \in E$ and $x \in \mathbb{R}$ such that

$$a < x < b \text{ and } x \notin E.$$

Let A, B be sets defined as

- $A := E \cap (-\infty, x) \subseteq E$.
- $B := E \cap (x, \infty) \subseteq E$.

Then we have

- $A \neq \varnothing$ and $B \neq \varnothing$ since $a \in A$ and $b \in B$.
- $\overline{A} \cap B = A \cap \overline{B} = \varnothing$.
- $E = A \cup B$ since $x \notin E$.

Thus A and B are separated, which is a contradiction. Therefore, we have

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

($\Longleftarrow$) Let $E \subseteq \mathbb{R}$ be a set such that

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

Assume that E is disconnected. Then there exist separated sets $A, B \subseteq \mathbb{R}$ such that

- $\varnothing \neq A \subseteq E$ and $\varnothing \neq B \subseteq E$.

- $\overline{A} \cap B = A \cap \overline{B} = \varnothing$.
- $E = A \cup B$.

Let $a \in A$ and $b \in B$ be given. Without loss of generality, assume that $a < b$. Let

$$\alpha := \sup(A \cap [a, b]) \in \mathbb{R},$$

where we have

- $a \in A \implies A \cap [a, b] \neq \varnothing$.
- $A \cap [a, b]$ is bounded above by b.

Then the *existence* of α is guaranteed by the *least upper bound property*. By Theorem 2.4.17, we have

$$\alpha \in \overline{(A \cap [a, b])} = (A \cap [a, b]) \cup (A \cap [a, b])'.$$

By Definition 2.4.4, it is clear that

$$(A \cap [a, b])' \subseteq A'.$$

Thus we have

$$\alpha \in (A \cap [a, b]) \subseteq A \quad \text{or} \quad \alpha \in (A \cap [a, b])' \subseteq A' \implies \alpha \in \overline{A}.$$

Then the following statements hold:

- $\alpha = \sup(A \cap [a, b]) \implies a \leq \alpha \leq b$.
- $\alpha \in \overline{A} \implies \alpha \notin B \implies \alpha \neq b$.

If $\alpha \notin A$, then we have

- $\alpha \neq a \implies a < \alpha < b$
- $\alpha \notin A$ and $\alpha \notin B \implies \alpha \notin E$

which is a contradiction. Otherwise, we have

$$\alpha \in A \implies \alpha \notin \overline{B}.$$

Then there exists $r \in (0, b - \alpha)$ such that

$$\widetilde{N}_r(\alpha) \cap B = \varnothing \implies \beta := \alpha + r \notin B.$$

Then we have

- $a \leq \alpha < \beta < b$
- $\beta > \alpha \implies \beta \notin A \implies \beta \notin E$

which is a contradiction. Therefore, we conclude that E is connected. $\square$

Chapter 3

Sequences and Series

3.1 Convergent Sequences

Let (X, d) be a metric space. A **sequence** $(x_n)_{n=1}^{\infty}$ is a function $f : \mathbb{N} \to X$, where $x_n = f(n)$ for each $n \in \mathbb{N}$.

A sequence $(x_n)_{n=1}^{\infty}$ is said to be **convergent** if there exists $x \in X$ such that

$$\text{For every } \varepsilon \in \mathbb{R}^+, \text{ there exists } N \in \mathbb{N} \text{ such that } n \geq N \implies d(x_n, x) < \varepsilon.$$

In this case, we say that

- $(x_n)_{n=1}^{\infty}$ **converges** to x.
- x is the **limit** of $(x_n)_{n=1}^{\infty}$.

We denote the limit of $(x_n)_{n=1}^{\infty}$ by

- $\lim_{n\to\infty} x_n = x$.
- $x_n \to x$ as $n \to \infty$.

If $(x_n)_{n=1}^{\infty}$ is not convergent, it is said to be **divergent**.

Example 3.1.1. Let $(\mathbb{R}, \|\cdot\|)$ be given. Let $(x_n)_{n=1}^{\infty}$ be a sequence defined as

$$x_n = \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

Then, for each $\varepsilon \in \mathbb{R}^+$, we have

$$n > \frac{1}{\varepsilon} \implies \|x_n - 0\| = \left|\frac{1}{n} - 0\right| = \frac{1}{n} < \varepsilon.$$

Therefore, $(x_n)_{n=1}^{\infty}$ converges to 0.

Let $\left(\mathbb{R}^+, \|\cdot\|\right)$ be a subspace of $(\mathbb{R}, \|\cdot\|)$. Let $(x_n)_{n=1}^{\infty}$ be a sequence defined as

$$x_n = \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

By Theorem 1.6.1, for each $y \in \mathbb{R}^+$, there exists $N_y \in \mathbb{N}$ such that $N_y > \dfrac{1}{y}$. Let

$$\varepsilon_y := y - \frac{1}{N_y} \in \mathbb{R}^+ \quad \text{for each } y \in \mathbb{R}^+.$$

Then we have

$$n > N_y \implies \|x_n - y\| = \left|\frac{1}{n} - y\right| > \left|\frac{1}{N_y} - y\right| = \varepsilon_y.$$

Therefore, $(x_n)_{n=1}^{\infty}$ diverges in $\left(\mathbb{R}^+, \|\cdot\|\right)$.

Remark. Let (X, d) be a metric space. In this textbook, $(x_n)_{n=1}^{\infty}$ represents a sequence in (X, d), while $\{x_n\}_{n=1}^{\infty}$ represents a set defined as

$$\{x_n\}_{n=1}^{\infty} := \{x_n \mid n \in \mathbb{N}\} \subseteq X.$$

Definition 3.1.2. Let (X, d) be a metric space. A sequence $(x_n)_{n=1}^{\infty}$ is said to be **Bounded** if $\{x_n\}_{n=1}^{\infty}$ is bounded.

Theorem 3.1.3. *Let (X, d) be a metric space. Let $(x_n)_{n=1}^{\infty}$ be a convergent sequence. Then we have*

- ***(A)*** *$(x_n)_{n=1}^{\infty}$ is bounded.*
- ***(B)*** *The limit of $(x_n)_{n=1}^{\infty}$ is unique.*

Proof. Let x be the limit of $(x_n)_{n=1}^{\infty}$.

(A) Let $\varepsilon = 1$. Then there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies d(x_n, x) < 1.$$

Let S be a set defined as

$$S := \{d(x_n, x) \mid 1 \leq n \leq N\} \subseteq [0, \infty).$$

Let $f : \mathbb{N}^N \to S$ be a function defined as

$$f(k) = d(x_k, x) \quad \text{for all } k \in \mathbb{N}^N.$$

Then f is surjective. By Lemma 2.2.3, S is finite. Then, by Theorem 2.2.5, S has a greatest element, say $M = \max S$. Then we have

$$d(x_n, x) < M + 1 \quad \text{for all } n \in \mathbb{N}.$$

Therefore, $(x_n)_{n=1}^{\infty}$ is bounded.

(B) Let y be the limit of $(x_n)_{n=1}^{\infty}$. Assume that $x \neq y$. Then we have $r := d(x,y) > 0$. Let $\varepsilon = \dfrac{r}{2}$. Then we have

- There exists $N_x \in \mathbb{N}$ such that $n \geq N_x \implies d(x_n, x) < \dfrac{r}{2}$.
- There exists $N_y \in \mathbb{N}$ such that $n \geq N_y \implies d(x_n, y) < \dfrac{r}{2}$.

Thus we have

$$n \geq \max\{N_x, N_y\} \implies d(x,y) \leq d(x,x_n) + d(x_n,y) < r,$$

which is a contradiction. Therefore, we have $x = y$.

□

Theorem 3.1.4. *Let (X,d) be a metric space. Then the following statements hold:*

(A) *A sequence $(x_n)_{n=1}^{\infty}$ converges to $x \in X$ if and only if*

For every $\varepsilon \in \mathbb{R}^+$, $N_\varepsilon(x)$ contains all but finitely many terms of $(x_n)_{n=1}^{\infty}$.

(B) *Let $E \subseteq X$ be given. If x is a limit point of E, then there exists a sequence $(x_n)_{n=1}^{\infty}$ in (X,d) such that*

$$\lim_{n\to\infty} x_n = x.$$

Proof. **(A)** **($\implies$)** Let $(x_n)_{n=1}^{\infty}$ converge to $x \in X$. Then, for each $\varepsilon \in \mathbb{R}^+$, there exists $M_\varepsilon \in \mathbb{N}$ such that

$$n \geq M_\varepsilon \implies d(x_n, x) < \varepsilon,$$

which implies that

$$\{x_n\}_{n=M_\varepsilon}^{\infty} \subseteq N_\varepsilon(x).$$

($\impliedby$) Let $(x_n)_{n=1}^{\infty}$ be a sequence in (X,d) such that

For every $\varepsilon \in \mathbb{R}^+$, $N_\varepsilon(x)$ contains all but finitely many terms of $(x_n)_{n=1}^{\infty}$.

(B) Let $E \subseteq X$ be given. Assume that x is a limit point of E. Then for every $\varepsilon \in \mathbb{R}^+$, we have $(N_\varepsilon(x) - \{x\}) \cap E \neq \varnothing$. Thus we can choose a sequence $(y_n)_{n=1}^{\infty}$ in (X,d) such that

$$y_n \in (N_{\frac{1}{n}}(x) \setminus \{x\}) \cap E.$$

Then we have

$$d(y_n, x) < \frac{1}{n} \quad \text{for each } n \in \mathbb{N},$$

which implies that $\lim\limits_{n\to\infty} y_n = x$.

□

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Appendix: Construction of $\mathbb{R}$

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